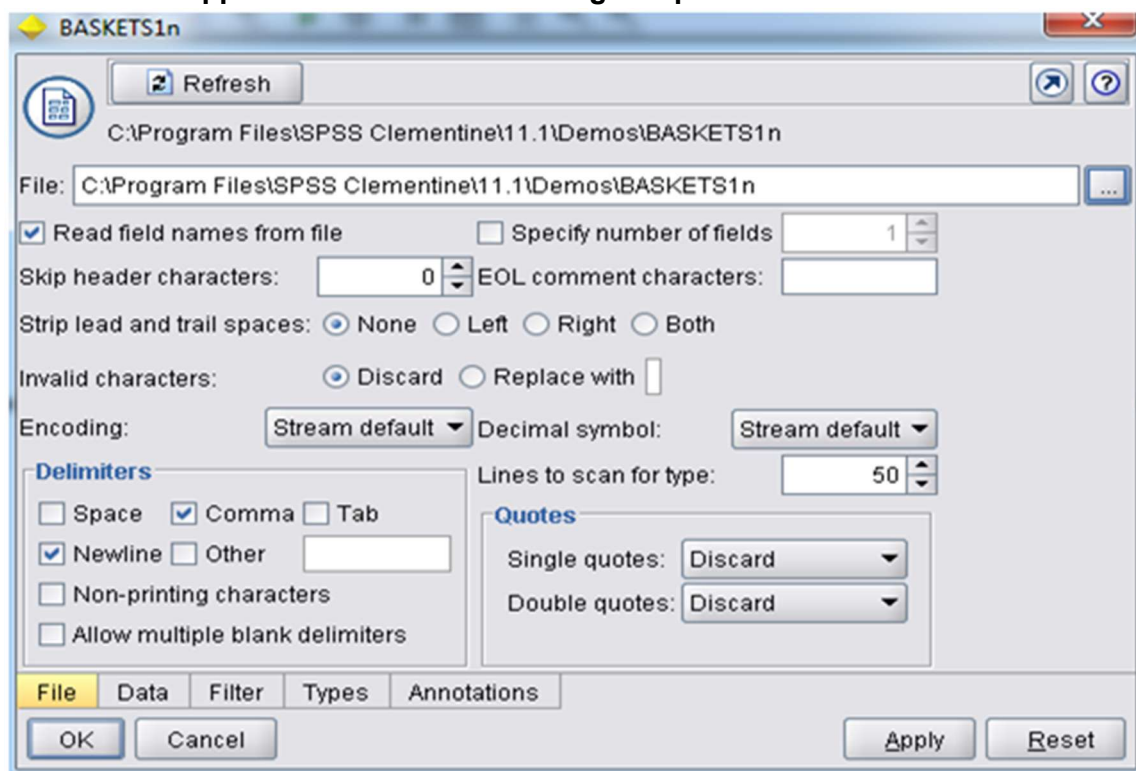


DATA MINING - CLEMENTINE RECORD

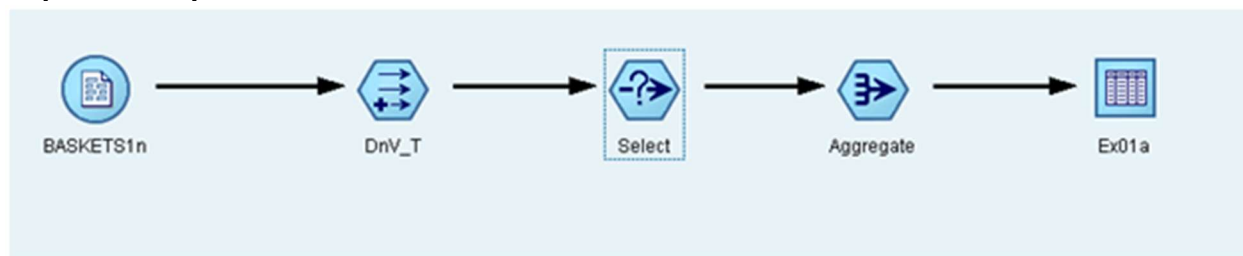
1. Using BASKETS1n dataset select the data as given below
 - a) Customer age < 35 and count the customers who buy dairy and VEG products
 - b) Find the AVG income of customers who buy atleast 5 products
 - c) Derive the field whose homeown is 'YES' and Age > 30 and sort data w.r.t. income in Ascending order, and output only the item fields.
 - d) Find the mean value of salary w.r.t age={Young, Middle, Senior}.

Input data set is applicable to all exercises in given problem statement: BASKETS1n

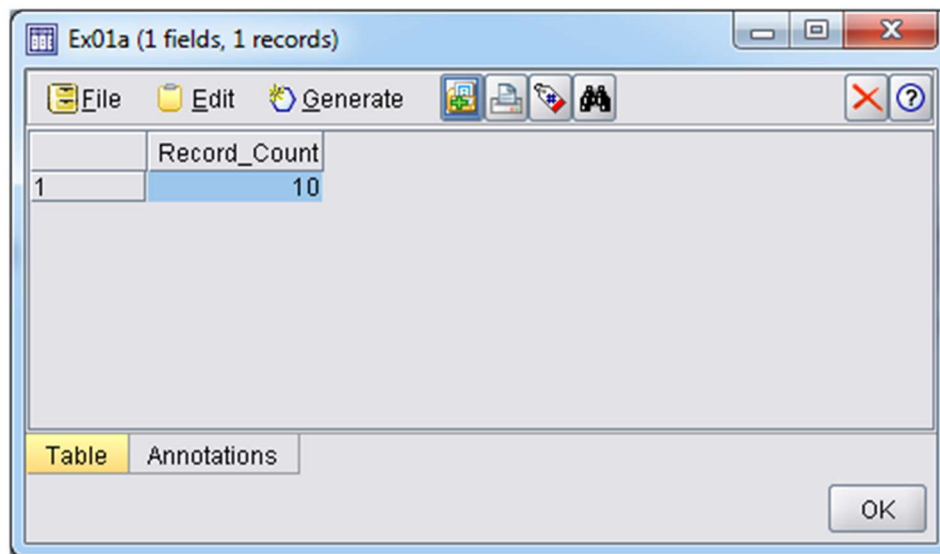


SOLUTION 1a)

Expected output:



Output dataset:

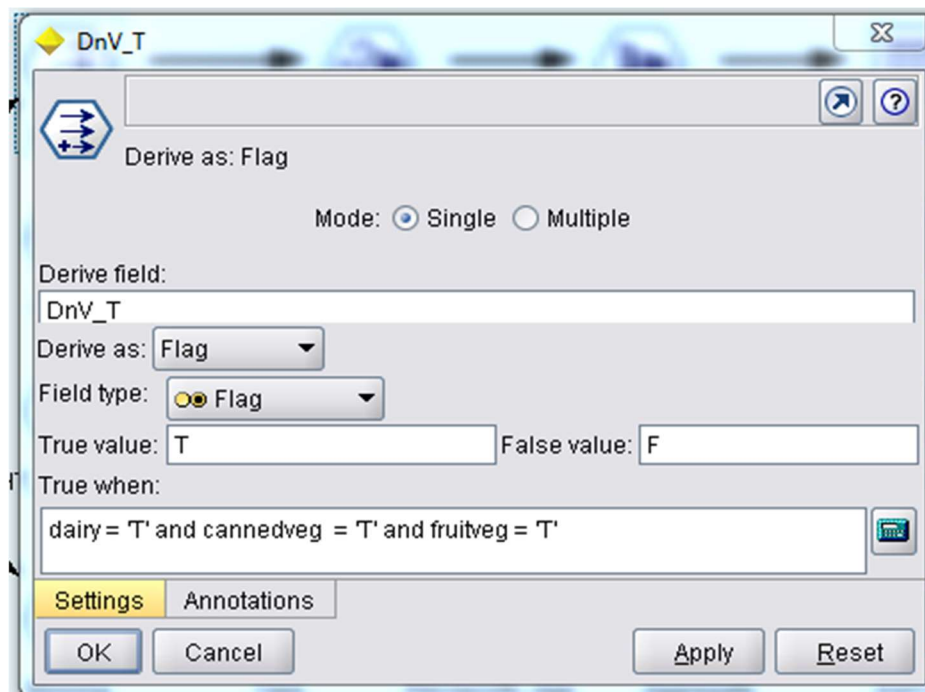


The screenshot shows a window titled 'Ex01a (1 fields, 1 records)'. It has a menu bar with 'File', 'Edit', and 'Generate'. Below the menu is a toolbar with icons for file operations. The main area contains a table with one column labeled 'Record_Count' and one row with the value '10'. At the bottom, there are tabs for 'Table' and 'Annotations', and an 'OK' button.

	Record_Count
1	10

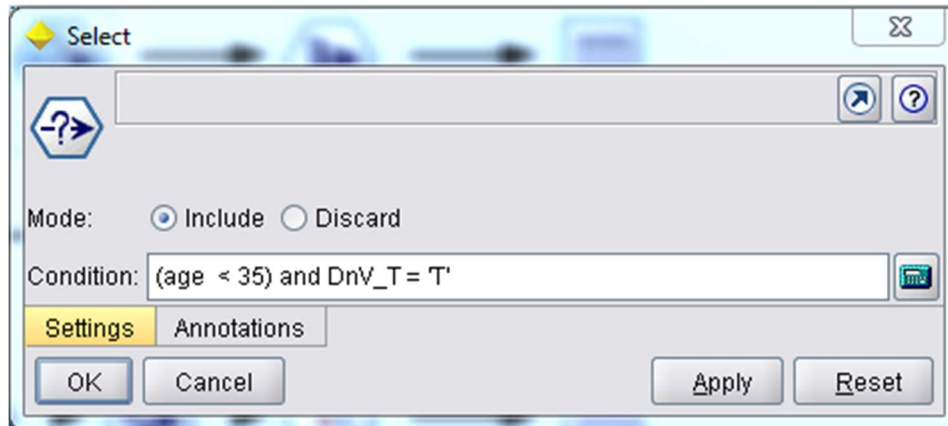
Procedure:

1. Specify the name of the file. You can enter a filename or click the ellipsis button (...) to select a file. The file path is shown once you have selected a file, and its contents are displayed with delimiters in the panel below it.
2. Select var.file from sources then goto C:\Program Files (x86)\SPSS Clementine\11.1\Demos\BASKETS1n we get the baskets in file.
3. Go to field options and select Derive flag and give condition as dairy = 'T' and cannedveg = 'T' and fruitveg = 'T' and click OK based on the conditions the truth values are shown and records are selected.

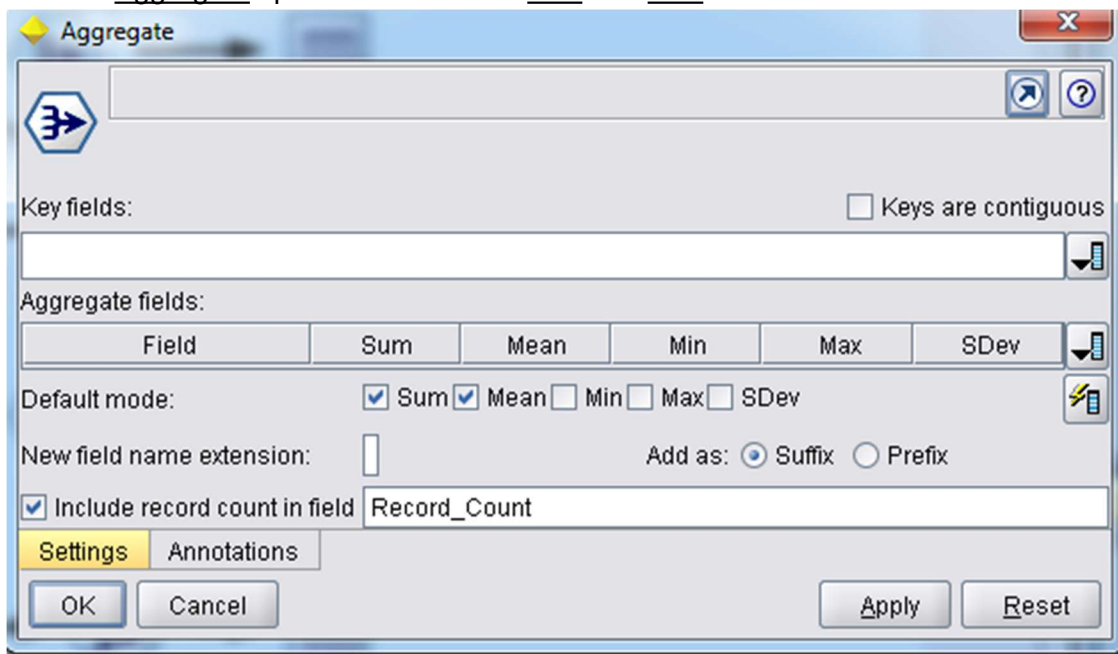


The screenshot shows the 'DnV_T' dialog box. It has a title bar with a yellow diamond icon and a close button. The main area is titled 'Derive as: Flag'. Below this, there are radio buttons for 'Mode: Single' (selected) and 'Multiple'. The 'Derive field:' section has a text box containing 'DnV_T'. The 'Derive as:' dropdown is set to 'Flag'. The 'Field type:' dropdown is set to 'Flag'. The 'True value:' text box contains 'T' and the 'False value:' text box contains 'F'. The 'True when:' text box contains the condition 'dairy = 'T' and cannedveg = 'T' and fruitveg = 'T''. At the bottom, there are tabs for 'Settings' and 'Annotations', and buttons for 'OK', 'Cancel', 'Apply', and 'Reset'.

4. Goto options search for the Select give the condition as [(age < 35) and DnV_T = 'T']



5. If both the conditions are true it counts the number of records.
6. Select the Aggregate operation to retrieve sum and max of the records.



7. Select output table and connect with aggregator
8. Right click on output table and execute it.
9. Then click on run now output will be displayed.

SOLUTION 1b)
Expected output:



Output dataset:

Ex01b (2 fields, 1 records)

	income_Mean	Record_Count
1	17911.806	144

Table Annotations

OK

Procedure:

1. Specify the name of the file. You can enter a filename or click the ellipsis button (...) to select a file. The file path is shown once you have selected a file, and its contents are displayed with delimiters in the panel below it.
2. Select var.file from sources then goto C:\Program Files (x86)\SPSS Clementine\11.1\Demos\BASKETS1n we get the baskets in file.
3. Goto options search for the Select give the condition as as shown in figure.

Select

Mode: ☒ Include ☐ Discard

Condition: `count_equal('T',[fruitveg,freshmeat,dairy,cannedveg,cannedmeat,frozenmeal,beer,wine,softdrink,fish,confectionery]) >= 5`

Settings Annotations

OK Cancel Apply Reset

4. If the condition is true it counts the number of records.
5. Select the Aggregate operation to retrieve income_Mean and count of the records.

Aggregate

Key fields: ☐ Keys are contiguous

Aggregate fields:

Field	Sum	Mean	Min	Max	SDev
income	☐	☒	☐	☐	☐

Default mode: ☐ Sum ☒ Mean ☐ Min ☐ Max ☐ SDev

New field name extension: Add as: ☒ Suffix ☐ Prefix

☒ Include record count in field

Settings Annotations

OK Cancel Apply Reset

6. Select output table and connect with aggregator
7. Right click on output table and execute it.
8. Then click on run now output will be displayed.

SOLUTION 1c)

Expected output:



Output dataset:

Ex01c (12 fields, 266 records)

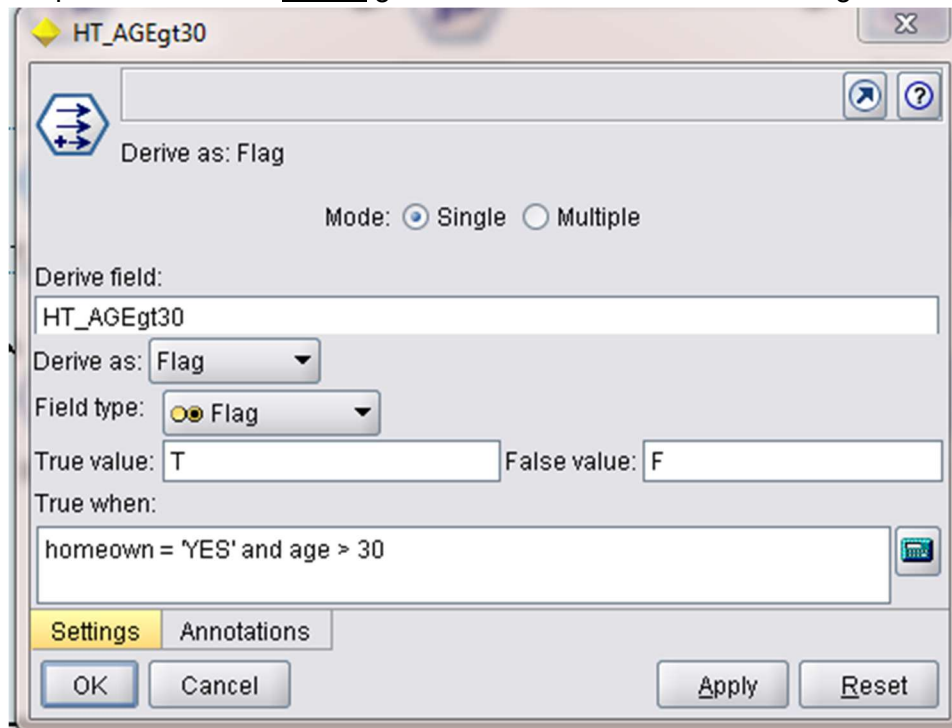
	fruitveg	freshmeat	dairy	cannedveg	cannedmeat	frozenmeal	beer	wine	softdrink	fish	confectionery	HT_AGEgt30
1	F	F	F	T	F	T	T	F	F	F	F	T
2	T	F	F	F	F	F	T	F	F	F	F	T
3	F	F	F	F	F	F	F	F	F	F	T	T
4	F	F	F	T	F	T	T	F	F	F	F	T
5	F	T	F	F	F	F	F	F	F	F	T	T
6	F	T	F	T	F	T	T	F	F	T	F	T

Table Annotations

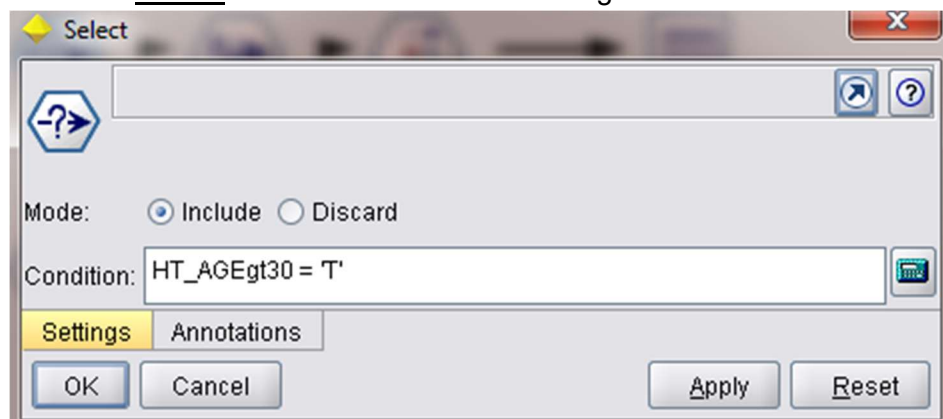
OK

Procedure:

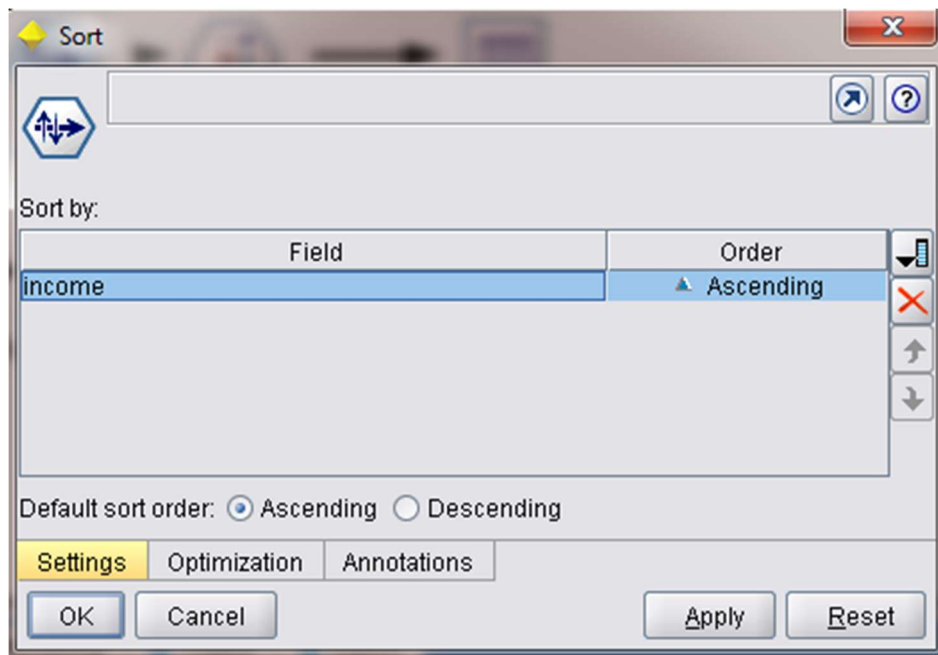
1. Specify the name of the file. You can enter a filename or click the ellipsis button (...) to select a file. The file path is shown once you have selected a file, and its contents are displayed with delimiters in the panel below it.
2. Select var.file from sources then goto C:\Program Files (x86)\SPSS Clementine\11.1\Demos\BASKETS1n we get the baskets in file.
3. Goto Field ops search for the Derive give the condition as as shown in figure.



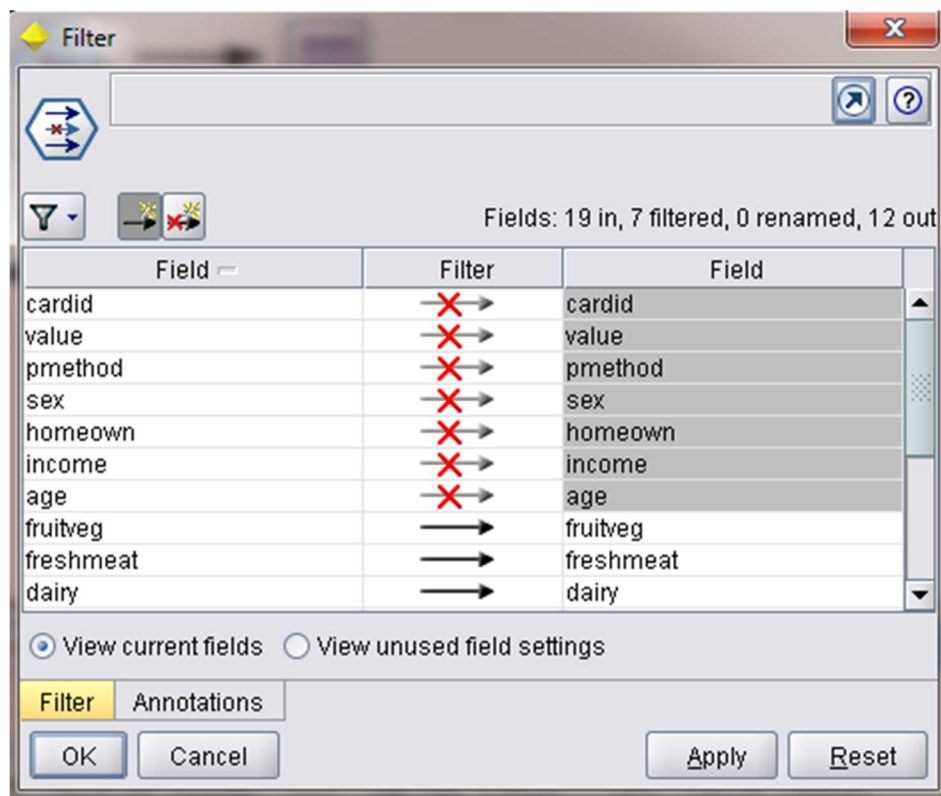
4. Connect Derive to Select with the condition shown in figure



5. Connect Select to Sort for sorting Income in ascending order



6. Connect Sort to Filter out all the non-item fields.



7. Select output table and connect with Filter.

7. Right click on output table and execute it.

8. Then click on run now output will be displayed.

SOLUTION 1d)

Expected output:



Output dataset:

Ex01d (2 fields, 3 records) #6

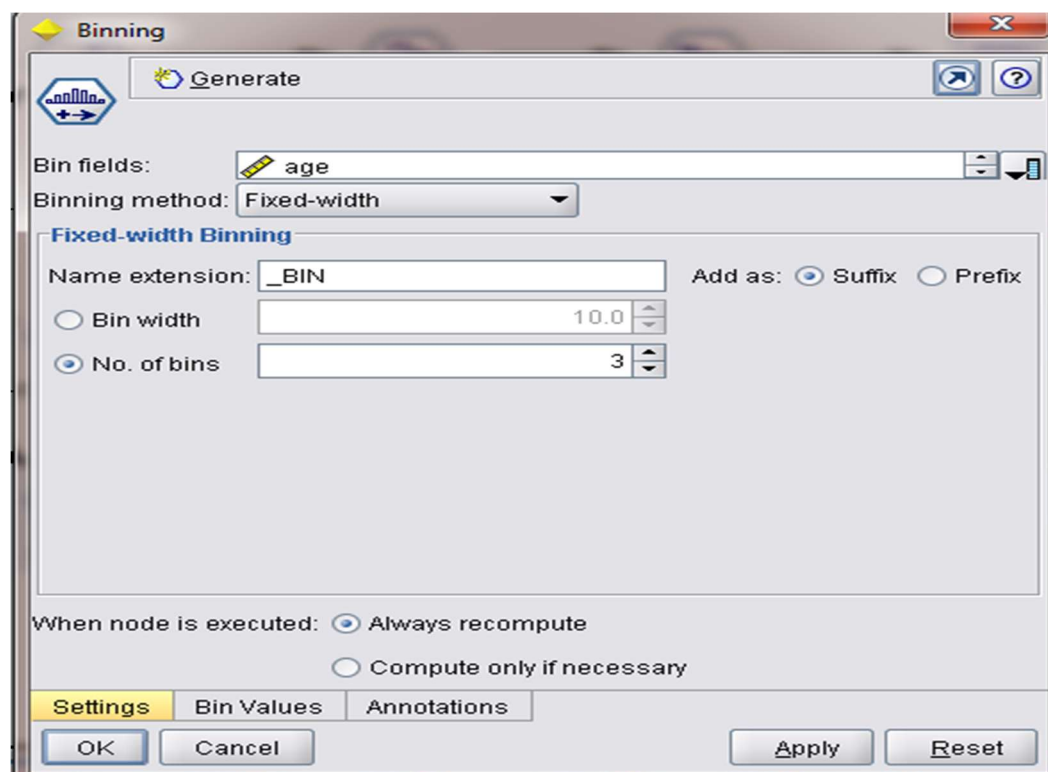
	income_Mean	Reclassify_Age
1	19868.852	Young
2	20517.869	Middle
3	20201.458	Senior

Table Annotations

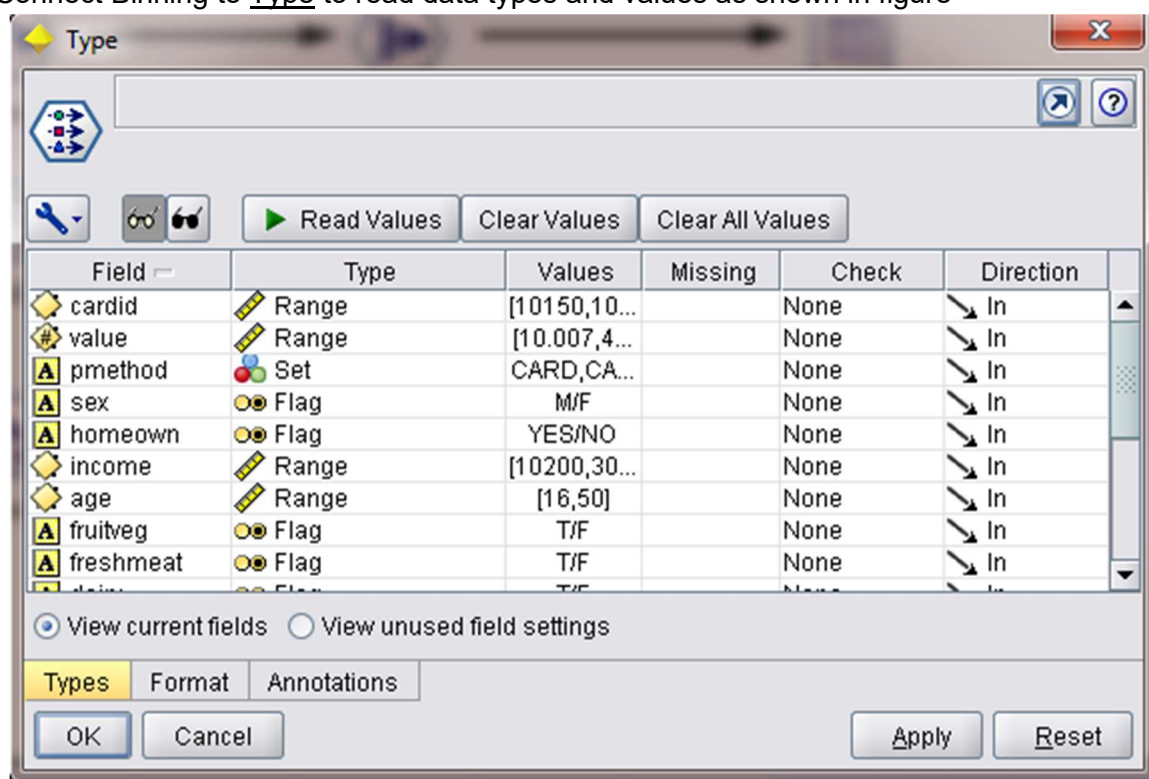
OK

Procedure:

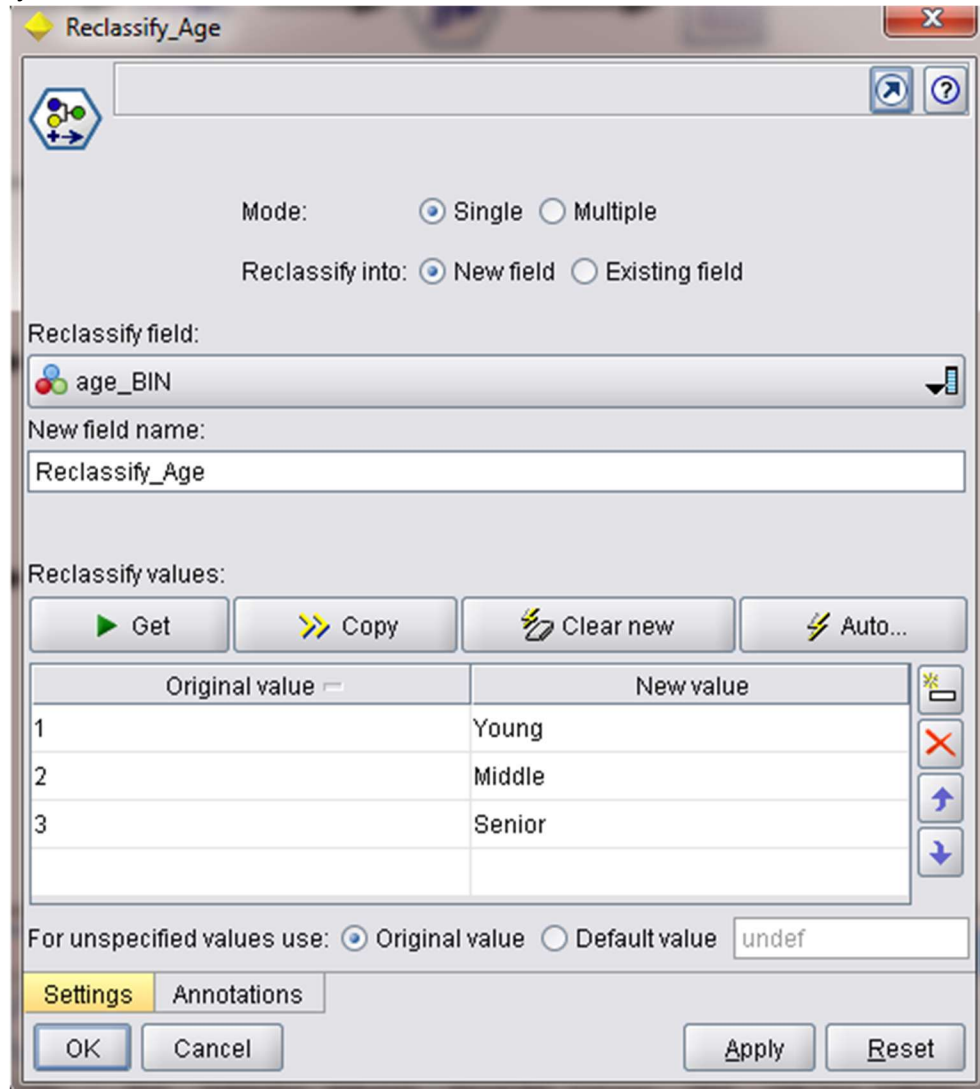
1. Specify the name of the file. You can enter a filename or click the ellipsis button (...) to select a file. The file path is shown once you have selected a file, and its contents are displayed with delimiters in the panel below it.
2. Select var.file from sources then goto C:\Program Files (x86)\SPSS Clementine\11.1\Demos\BASKETS1n we get the baskets in file.
3. Goto Field ops search for the Binning give the condition as as shown in figure.



4. Connect Binning to Type to read data types and values as shown in figure



5. Connect Type to Reclassify, for classifying age binned in 1,2,3 to Young,Middle,Senior respectively.

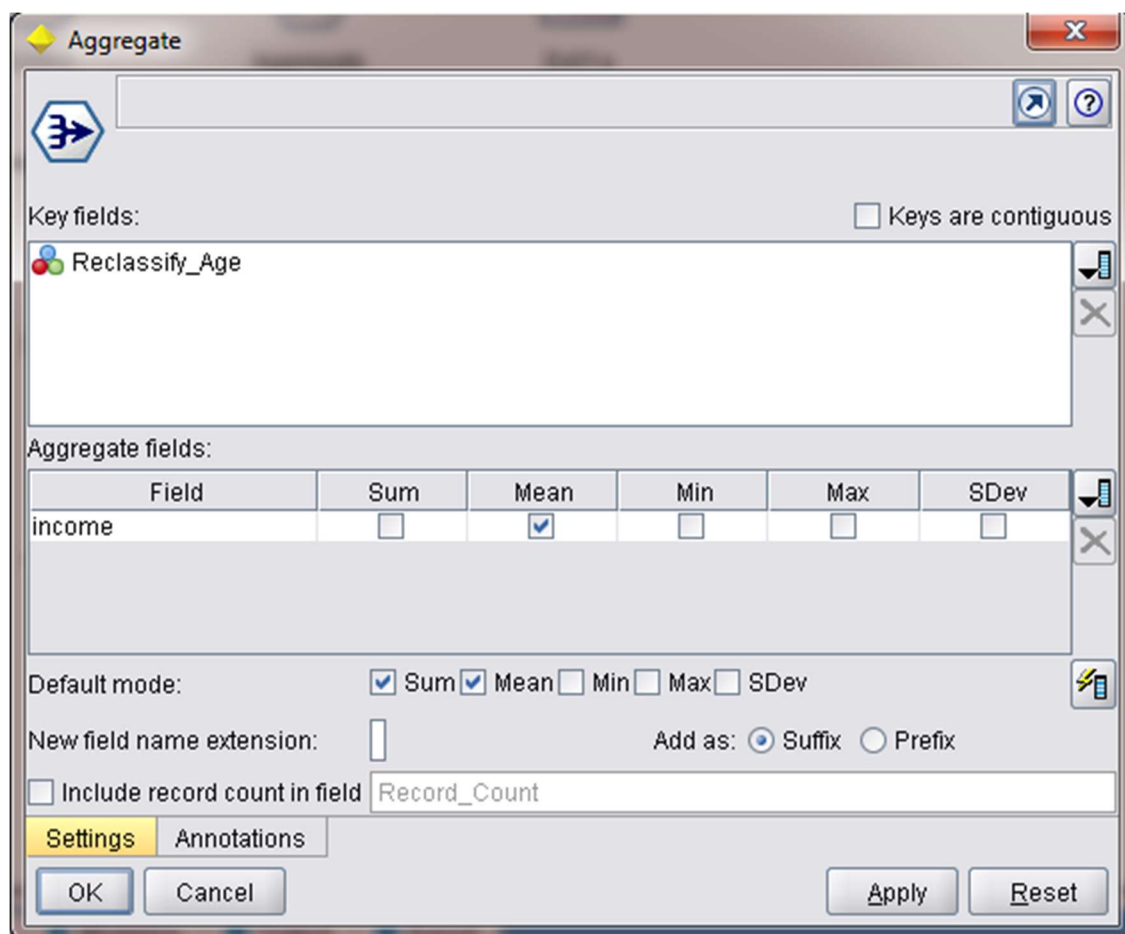


The image shows a software dialog box titled "Reclassify_Age". It has a standard window with a title bar and a close button. The dialog is divided into several sections:

- Mode:** Two radio buttons, "Single" (selected) and "Multiple".
- Reclassify into:** Two radio buttons, "New field" (selected) and "Existing field".
- Reclassify field:** A text box containing "age_BIN" with a dropdown arrow on the right.
- New field name:** A text box containing "Reclassify_Age".
- Reclassify values:** A section with four buttons: "Get" (green play icon), "Copy" (yellow double arrow icon), "Clear new" (lightning bolt icon), and "Auto..." (lightning bolt icon). Below these buttons is a table with two columns: "Original value" and "New value".
- For unspecified values use:** Two radio buttons, "Original value" (selected) and "Default value", followed by a text box containing "undef".
- Settings/Annotations:** Two tabs, "Settings" (active) and "Annotations".
- Buttons:** "OK", "Cancel", "Apply", and "Reset" at the bottom.

Original value	New value
1	Young
2	Middle
3	Senior

6. Connect Reclassify to Aggregate to get the income_Mean w.r.t. Different age categories.



7. Select output table and connect with Aggregate.
8. Right click on output table and execute it.
9. Then click on run now output will be displayed.

2. Using DRUG3n and DRUG4n datasets select the data as given below

a) Select 50% of records where maximum type of drug are present along with no restrictions on remaining drugs, and use histogram graph of age w.r.t BP

b) Take the equal number of samples of each drug and calculate the Std. Dev. of age w.r.t drug and compare it with complete data Std. Dev. of age w.r.t drug and give a conclusion statement.

c) List 5 strong associations of attribute values, and derive and display the data.

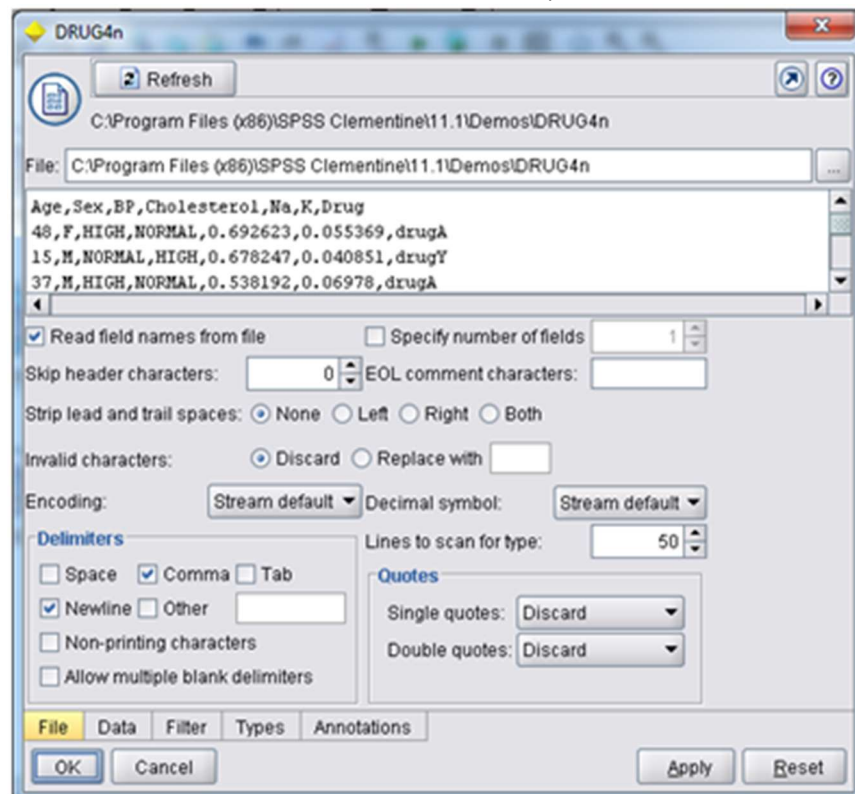
d) Append DRUG2n dataset to given datasets and consider distinct values of Age.

e) Using the above 3 datasets (DRUG2n, DRUG3n, DRUG4n) perform the following

i) Young_Age ≤ 30 , Middle_Age > 30 and ≤ 50 , Senior_Age > 50

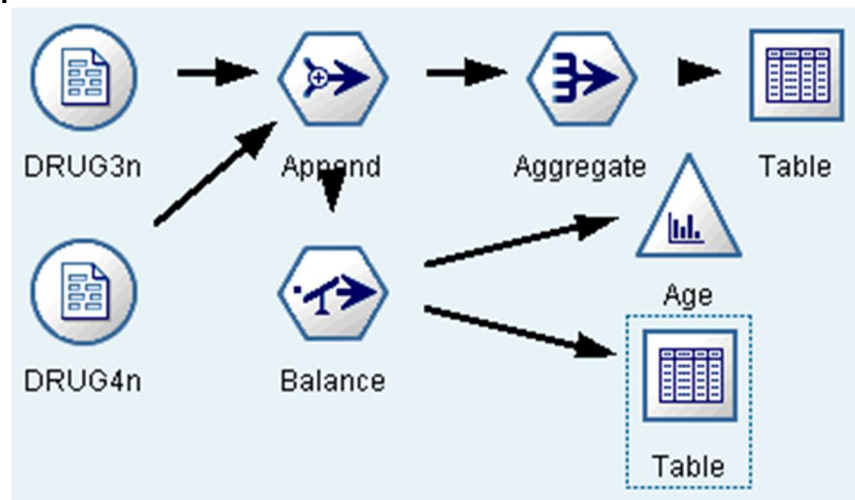
ii) Multi plot the above Age categories with Na and K and drug

Input data set is applicable to all exercises in given problem statement: DRUG3n and DRUG4n (For exercises d and e DRUG2n is also used)

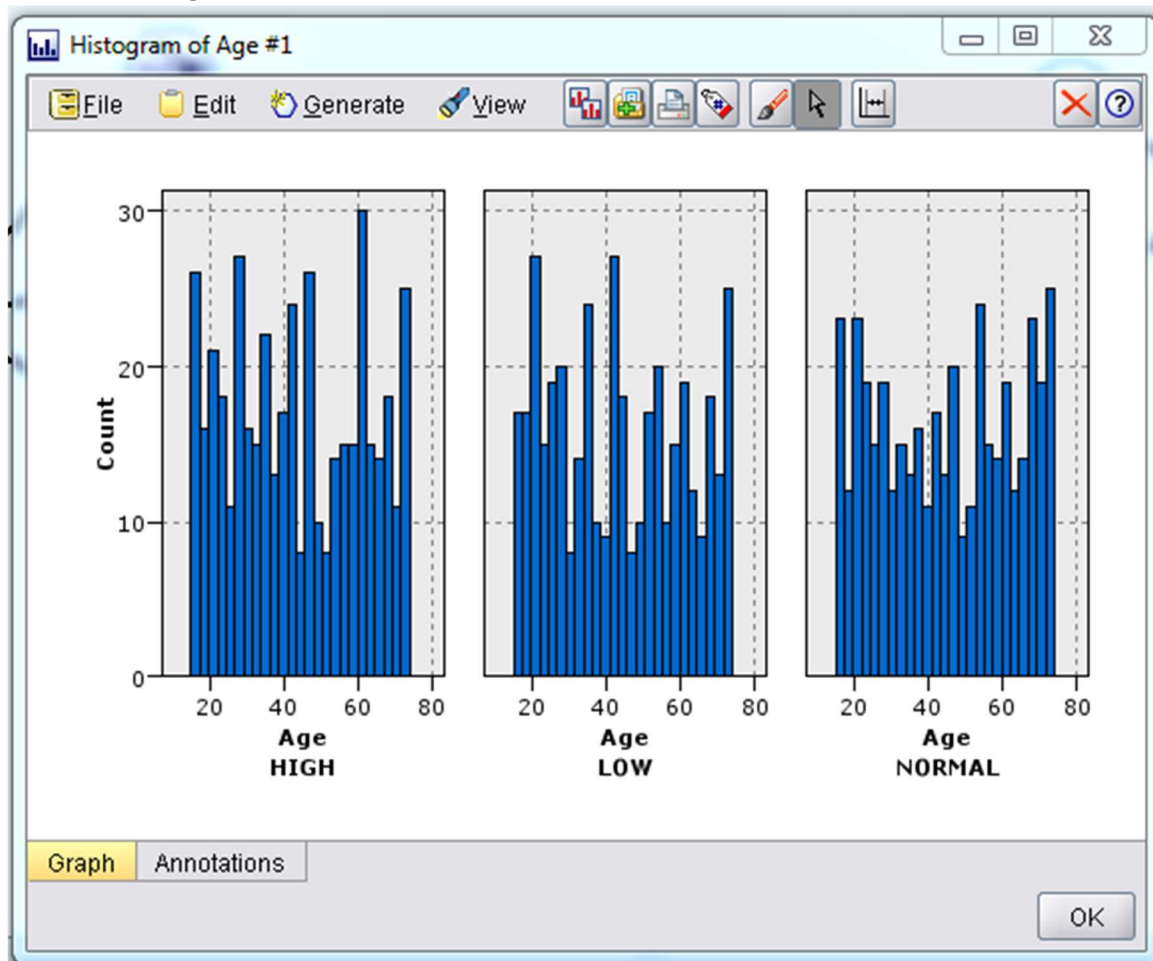


SOLUTION 2a)

Expected output:



Output dataset/ graph:



Procedure: The following are the nodes used for this exercise with respective settings.

APPEND:

Append

Append 2 datasets

Match fields by: ☐ Position ☒ Name ☐ Match case

Preview of field matches and structure

Output Field	1[DRUG3n:DRUG3n]	2[DRUG4n:DRUG4n]
Age	Age	Age
Sex	Sex	Sex
BP	BP	BP
Cholesterol	Cholesterol	Cholesterol
Na	Na	Na
K	K	K
Drug	Drug	Drug

Include fields from: ☒ Main dataset only ☐ All datasets

☐ Tag records by including source dataset in field

Inputs **Append** Annotations

OK Cancel Apply Reset

Aggregate: to get the count of all Drug types as shown in following output table.

Aggregate

Key fields: ☐ Keys are contiguous

Drug

Aggregate fields:

Field	Sum	Mean	Min	Max	SDev

Default mode: ☐ Sum ☐ Mean ☐ Min ☐ Max ☐ SDev

New field name extension: Add as: ☒ Suffix ☐ Prefix

☒ Include record count in field

Settings Annotations

OK Cancel Apply Reset

Table: from the following output we can identify that 'drugY' has maximum number of records when compared to remaining Drug types.

Table (2 fields, 5 records) #4

	Drug	Record_Count
1	drugB	121
2	drugX	426
3	drugC	153
4	drugY	719
5	drugA	181

Table Annotations

Balance: for selecting 50% of records for 'drugY'

Balance

Balance Directives:

Factor	Condition
0.5	Drug='drugY'

Settings Annotations

OK Cancel Apply Reset

Histogram: The following Histogram node gives the output of Age w.r.t. BP

Age

X: Age

Field: Age

Overlay

Color: Panel: BP Animation:

Plot Options Appearance Output Annotations

OK Execute Cancel Apply Reset

Table: It is an output of the records after selecting 50% of 'drugY' and no restrictions on remaining Drugs

Table (7 fields, 1,230 records)

FileEditGenerate

	Age	Sex	BP	Cholesterol	Na	K	Drug
1	25	F	HIGH	HIGH	0.676	0.075	drugA
2	17	F	HIGH	HIGH	0.540	0.030	drugY
3	74	F	LOW	HIGH	0.850	0.077	drugC
4	40	F	NORMAL	HIGH	0.677	0.050	drugX
5	32	F	HIGH	HIGH	0.582	0.025	drugY
6	64	M	HIGH	NORMAL	0.641	0.078	drugB
7	45	M	HIGH	HIGH	0.664	0.048	drugA
8	33	F	LOW	NORMAL	0.822	0.028	drugY
9	73	M	HIGH	NORMAL	0.792	0.062	drugB
10	27	F	HIGH	NORMAL	0.555	0.047	drugA
11	62	M	HIGH	NORMAL	0.510	0.071	drugB
12	72	M	HIGH	NORMAL	0.819	0.074	drugB
13	19	M	HIGH	NORMAL	0.553	0.033	drugY
14	28	M	HIGH	HIGH	0.584	0.068	drugA
15	54	M	NORMAL	NORMAL	0.606	0.077	drugX
16	37	F	NORMAL	HIGH	0.744	0.021	drugY
17	33	F	NORMAL	NORMAL	0.815	0.065	drugX
18	73	F	LOW	HIGH	0.523	0.026	drugY
19	41	F	NORMAL	HIGH	0.797	0.075	drugX
20	41	M	LOW	HIGH	0.846	0.055	drugY

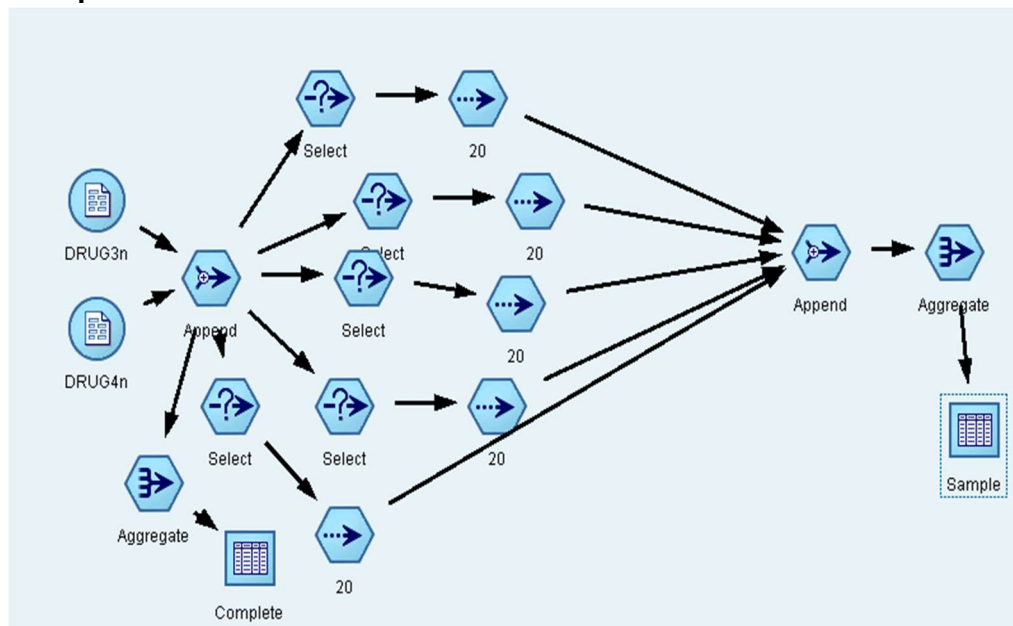
Table

Annotations

OK

SOLUTION 2b)

Expected output:



Output dataset:

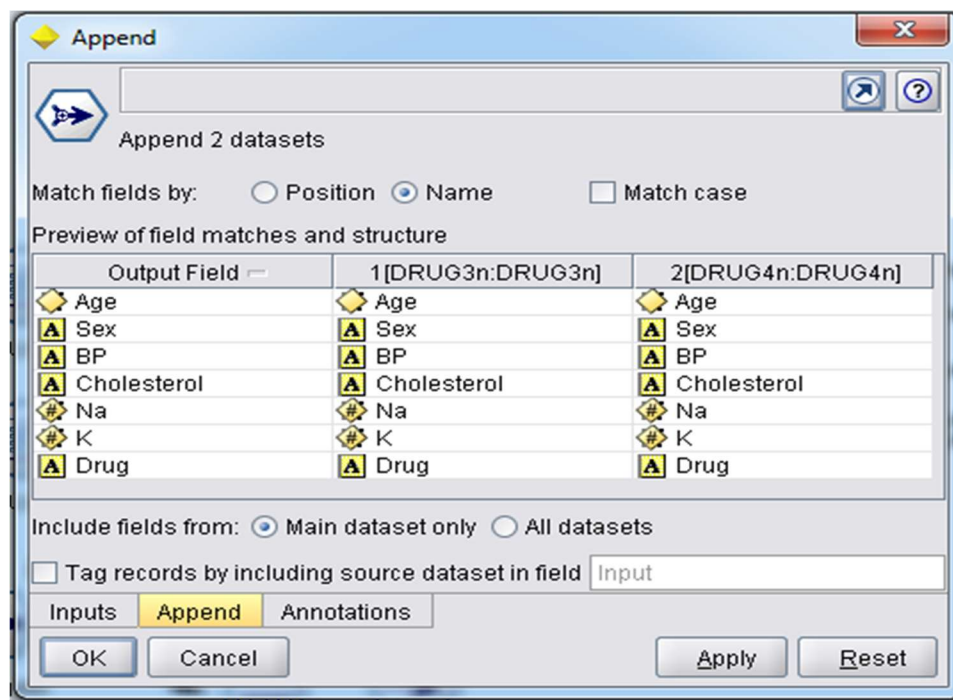
Sample (2 fields, 5 records) #1		
	Age_SDev	Drug
1	20.072	drugY
2	18.847	drugX
3	17.136	drugC
4	6.307	drugB
5	11.537	drugA

Complete (2 fields, 5 records)		
	Age_SDev	Drug
1	6.488	drugB
2	17.648	drugX
3	17.868	drugC
4	17.612	drugY
5	10.173	drugA

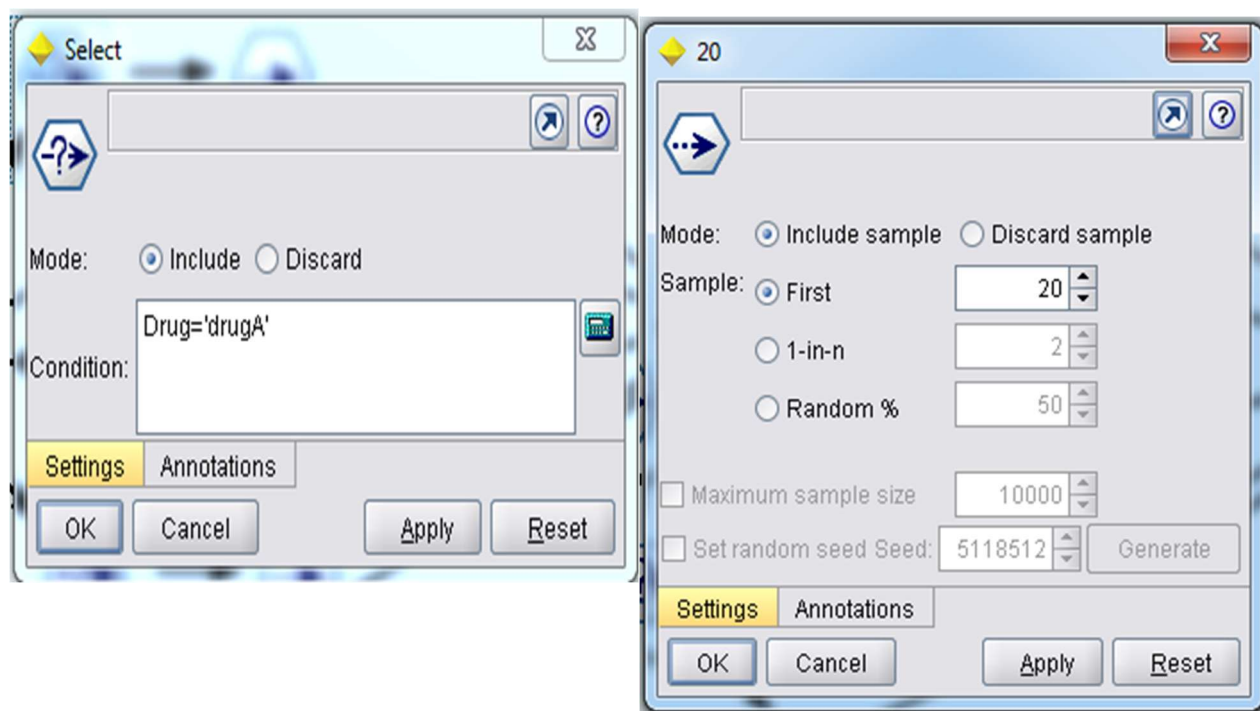
When the above Sample data output and complete data output is compared Standard Deviation of Age w.r.t Each drug type is almost similar, but there is a bit difference in Standard Deviation of Age w.r.t drugX, drugY and drugC in Sample data where as in complete data Standard Deviation of Age for the above drug types has minor difference.

Procedure: The following are the nodes used for this exercise with respective settings.

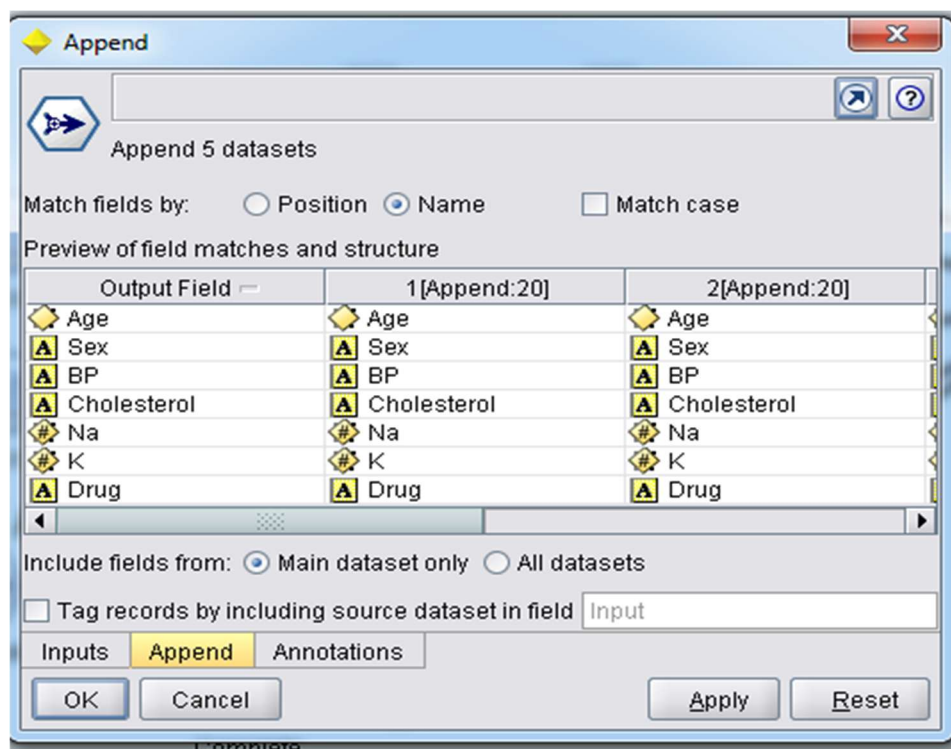
APPEND



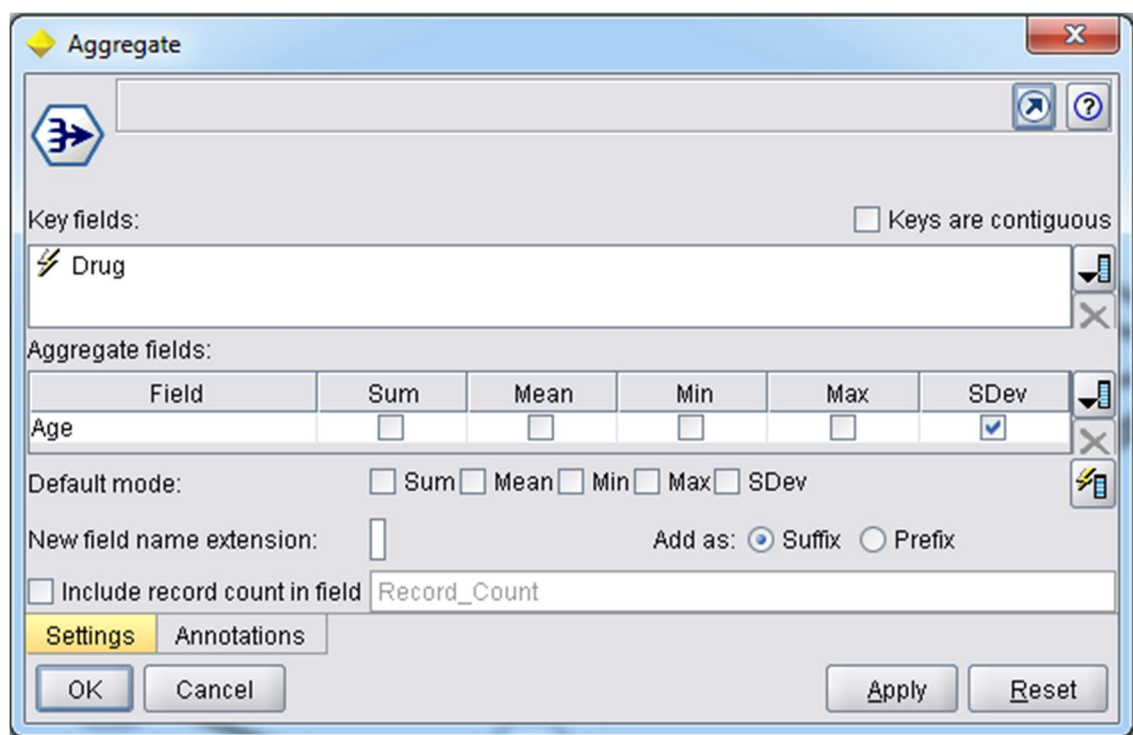
SELECT and SAMPLE: This procedure is followed for remaining drug types where 20 equal samples of each drug type is selected



APPEND: Appending all samples



Sample Aggregate: Aggregate of Sample data



Complete Aggregate: Aggregate of Complete data

Aggregate

Key fields: ☐ Keys are contiguous

Drug

Aggregate fields:

Field	Sum	Mean	Min	Max	SDev
	☒	☒	☐	☐	☐

Default mode: ☒ Sum ☒ Mean ☐ Min ☐ Max ☐ SDev

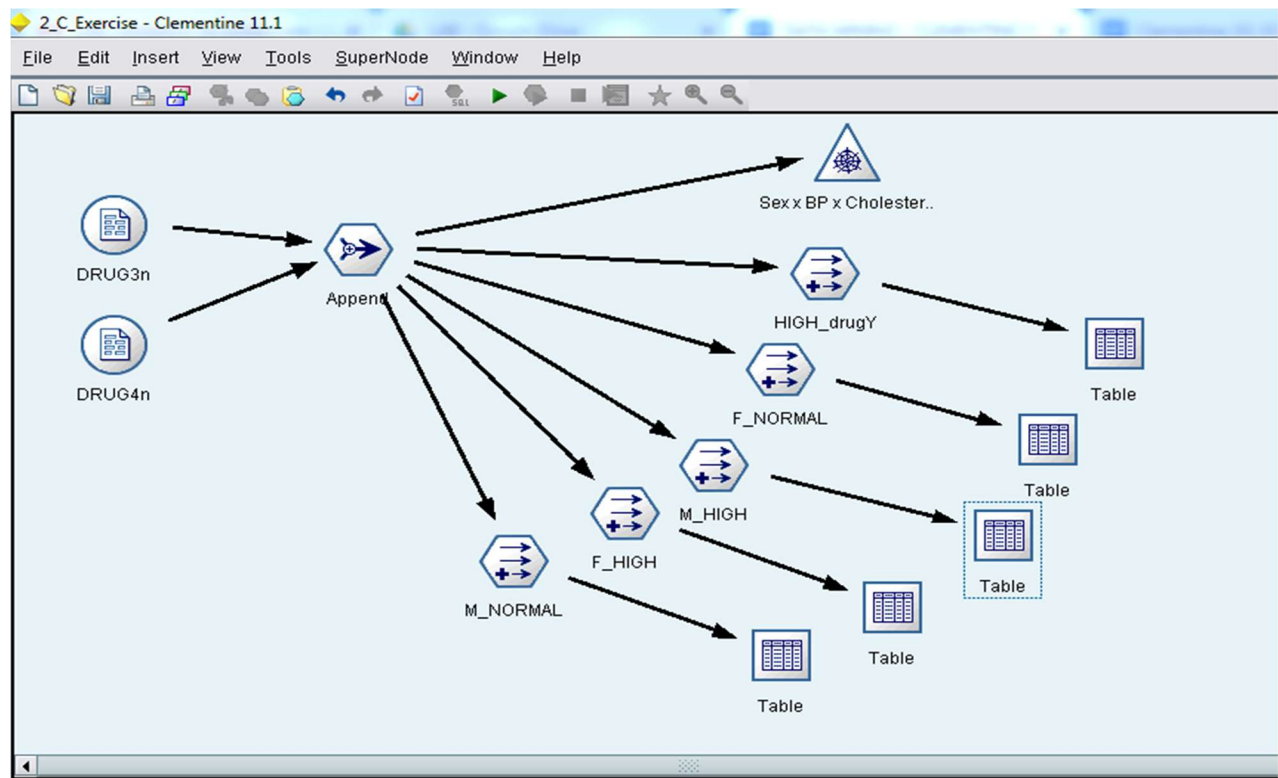
New field name extension: Add as: ☒ Suffix ☐ Prefix

☐ Include record count in field

Settings Annotations

OK Cancel Apply Reset

SOLUTION 2c)
Expected output:



Output dataset: The following is the output for Sex = 'M' and Cholesterol = 'High'

	Age	Sex	BP	Cholesterol	Na	K	Drug	M_HIGH
1	25	F	HIGH	HIGH	0.676	0.075	drugA	F
2	17	F	HIGH	HIGH	0.540	0.030	drugY	F
3	23	M	LOW	NORMAL	0.556	0.036	drugY	F
4	24	M	NORMAL	NORMAL	0.845	0.055	drugY	F
5	74	F	LOW	HIGH	0.850	0.077	drugC	F
6	40	F	NORMAL	HIGH	0.677	0.050	drugX	F
7	32	F	HIGH	HIGH	0.582	0.025	drugY	F
8	70	M	LOW	HIGH	0.716	0.037	drugY	T
9	64	M	HIGH	NORMAL	0.641	0.078	drugB	F
10	45	M	HIGH	HIGH	0.664	0.048	drugA	T
11	33	F	LOW	NORMAL	0.822	0.028	drugY	F
12	74	F	LOW	NORMAL	0.772	0.048	drugY	F
13	73	M	HIGH	NORMAL	0.792	0.062	drugB	F
14	38	F	LOW	HIGH	0.794	0.052	drugY	F
15	72	F	HIGH	NORMAL	0.534	0.021	drugY	F

Procedure: The following are the nodes used for this exercise with respective settings.

APPEND

Append

Append 2 datasets

Match fields by: ☐ Position ☒ Name ☐ Match case

Preview of field matches and structure

Output Field	1[DRUG3n:DRUG3n]	2[DRUG4n:DRUG4n]
Age	Age	Age
Sex	Sex	Sex
BP	BP	BP
Cholesterol	Cholesterol	Cholesterol
Na	Na	Na
K	K	K
Drug	Drug	Drug

Include fields from: ☒ Main dataset only ☐ All datasets

☐ Tag records by including source dataset in field

Inputs **Append** Annotations

OK Cancel Apply Reset

WEB: Plotting the web for Sex, BP, Cholesterol and Drug to get 5 strong associations

Sex x BP x Cholesterol x Drug

Threshold values are: Absolute, Strong links are heavier

☒ Web ☐ Directed web

Fields:

- Sex
- BP
- Cholesterol
- Drug

☐ Show true flags only

Line values are: Absolute

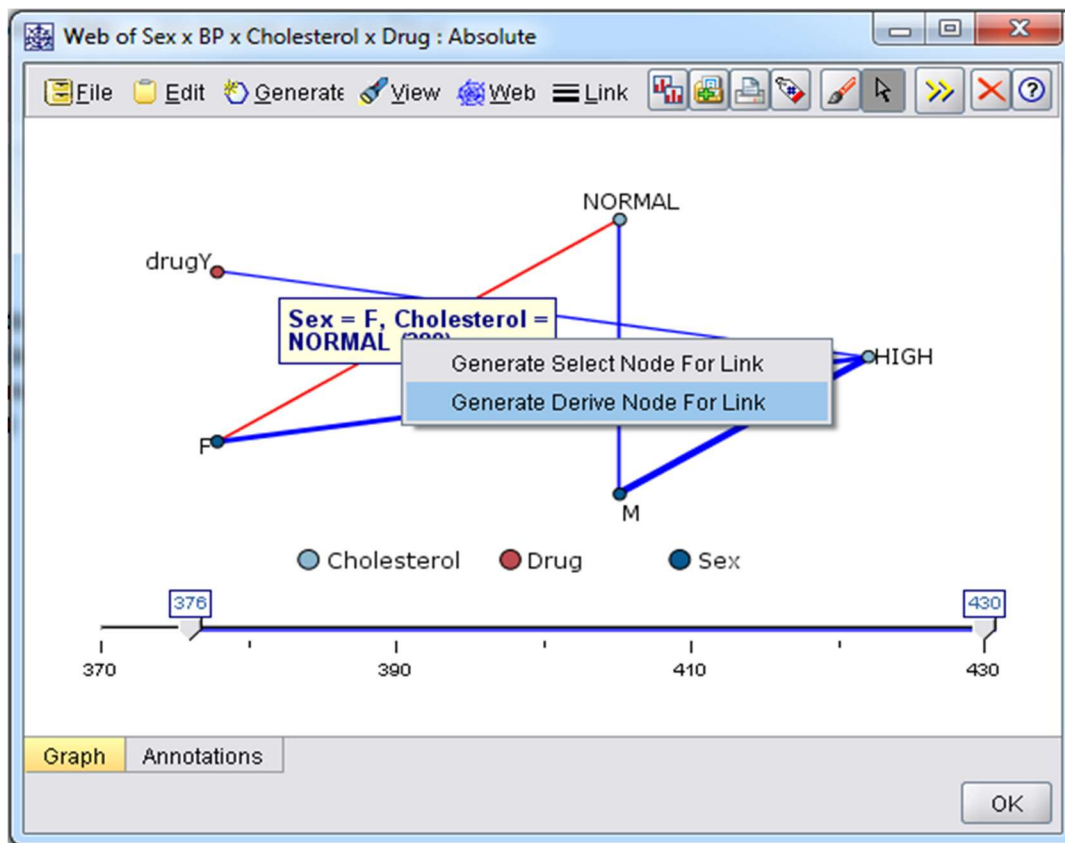
☒ Strong links are heavier

☐ Weak links are heavier

Plot Options Appearance Output Annotations

OK Execute Cancel Apply Reset

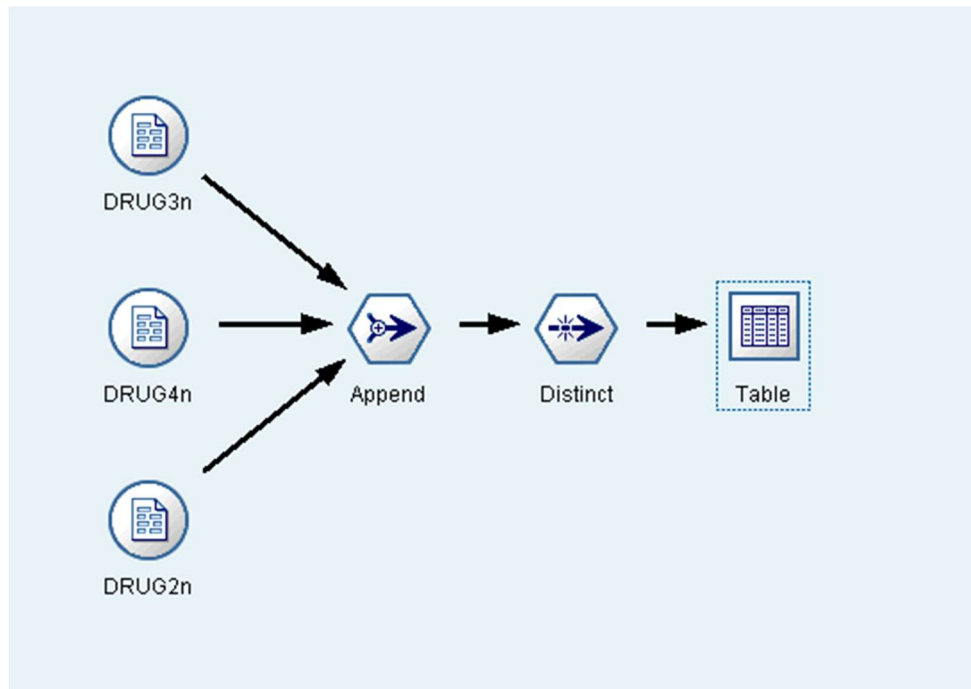
When web is created showing 5 strong links, we have to derive nodes for every links by right clicking on link and generate derive node for link.



When a derive node is created from link the following is the configuration of derive node

SOLUTION 2d)

Expected output:

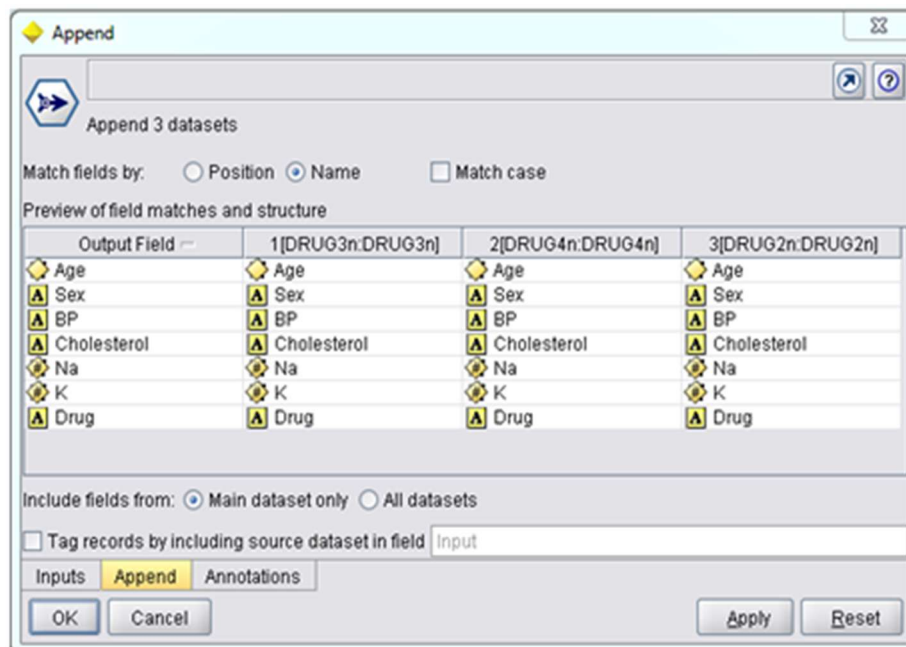


Output dataset: The following output is showing the records with distinct value of ages.

	Age	Sex	BP	Cholesterol	Na	K	Drug
1	25	F	HIGH	HIGH	0.676	0.075	drugA
2	17	F	HIGH	HIGH	0.540	0.030	drugY
3	23	M	LOW	NORMAL	0.556	0.036	drugY
4	24	M	NORMAL	NORMAL	0.845	0.055	drugY
5	74	F	LOW	HIGH	0.850	0.077	drugC
6	40	F	NORMAL	HIGH	0.677	0.050	drugX
7	32	F	HIGH	HIGH	0.582	0.025	drugY
8	70	M	LOW	HIGH	0.716	0.037	drugY
9	64	M	HIGH	NORMAL	0.641	0.078	drugB
10	45	M	HIGH	HIGH	0.664	0.048	drugA
11	33	F	LOW	NORMAL	0.822	0.028	drugY
12	73	M	HIGH	NORMAL	0.792	0.062	drugB
13	38	F	LOW	HIGH	0.794	0.052	drugY
14	72	F	HIGH	NORMAL	0.534	0.021	drugY
15	27	F	HIGH	NORMAL	0.555	0.047	drugA
16	62	M	HIGH	NORMAL	0.510	0.071	drugB
17	19	M	HIGH	NORMAL	0.553	0.033	drugY
18	28	M	HIGH	HIGH	0.584	0.068	drugA
19	54	M	NORMAL	NORMAL	0.606	0.077	drugX
20	37	F	NORMAL	HIGH	0.744	0.021	drugY

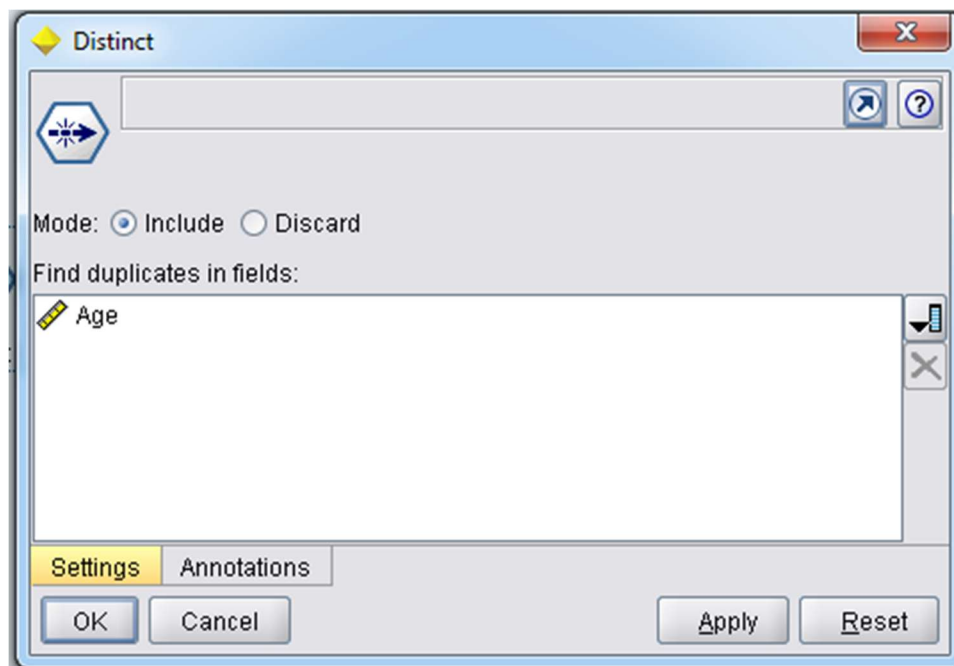
Procedure: The following are the nodes used for this exercise with respective settings.

Append:



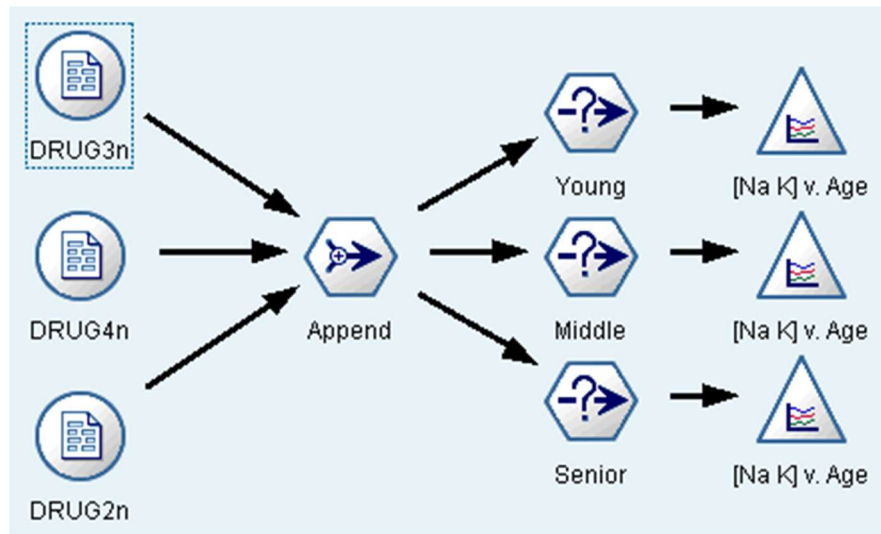
Appended data sets are given in accordance with age are exported to output graphs.

Distinct Age:

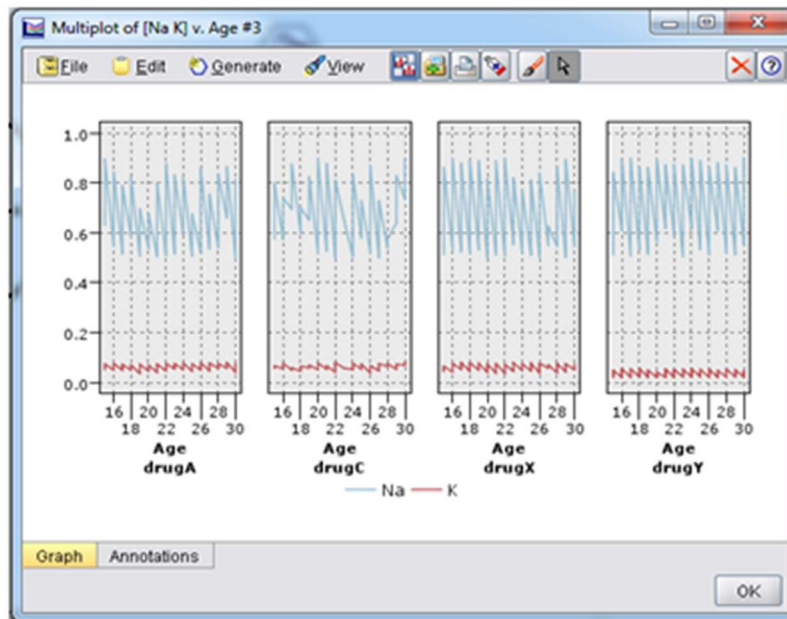


SOLUTION 2e)

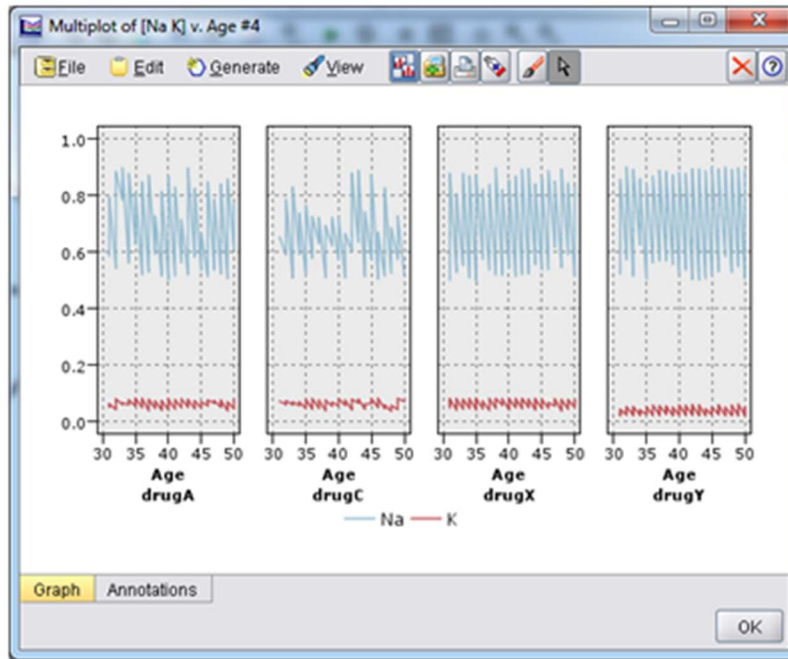
Expected output:



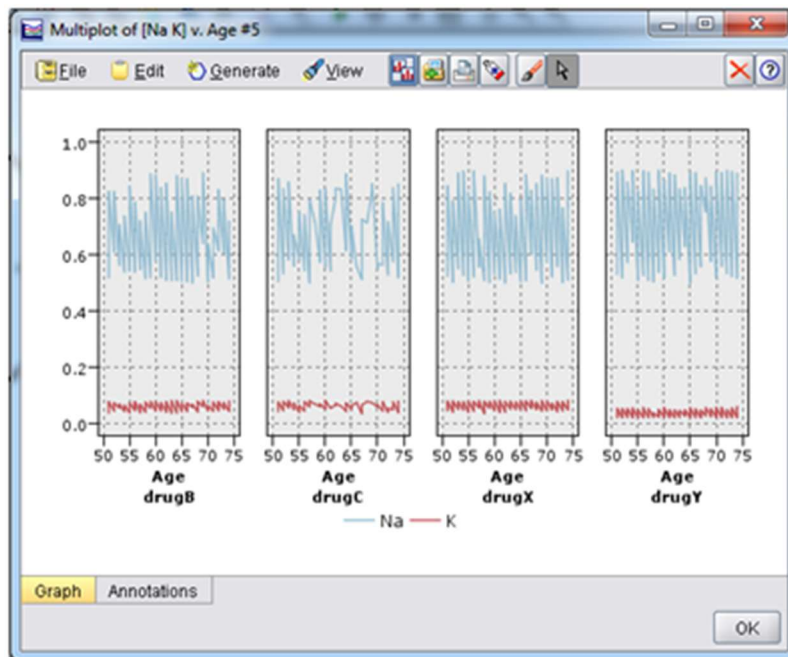
Output dataset/graph:



Above is Multi plot the above Young_Age categories with Na and K and drug



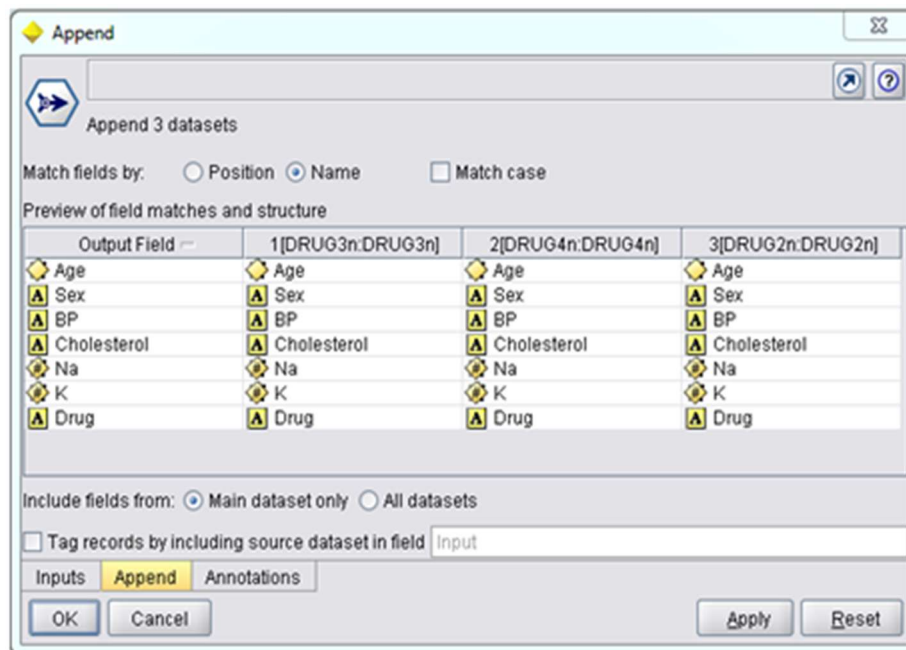
Above is Multi plot the above Middle_Age categories with Na and K and drug



Above is Multi plot the above Senior_Age categories with Na and K and drug

Procedure: The following are the nodes used for this exercise with respective settings.

Append:

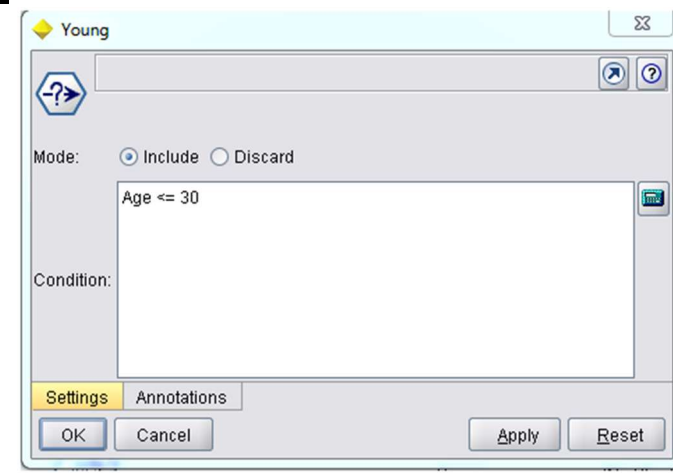


The 'Append' dialog box is shown with the title 'Append'. It contains a section 'Append 3 datasets'. Below this, there are radio buttons for 'Match fields by: Position', 'Name' (selected), and a checkbox for 'Match case'. A section titled 'Preview of field matches and structure' contains a table with four columns: 'Output Field', '1[DRUG3n:DRUG3n]', '2[DRUG4n:DRUG4n]', and '3[DRUG2n:DRUG2n]'. The table lists fields: Age, Sex, BP, Cholesterol, Na, K, and Drug, each with a yellow triangle icon. Below the table, there are radio buttons for 'Include fields from: Main dataset only' (selected) and 'All datasets'. A checkbox 'Tag records by including source dataset in field' is followed by a text input field containing 'Input'. At the bottom, there are tabs for 'Inputs', 'Append' (selected), and 'Annotations', and buttons for 'OK', 'Cancel', 'Apply', and 'Reset'.

Output Field	1[DRUG3n:DRUG3n]	2[DRUG4n:DRUG4n]	3[DRUG2n:DRUG2n]
Age	Age	Age	Age
Sex	Sex	Sex	Sex
BP	BP	BP	BP
Cholesterol	Cholesterol	Cholesterol	Cholesterol
Na	Na	Na	Na
K	K	K	K
Drug	Drug	Drug	Drug

Appended data sets are given in accordance with age are exported to output graphs.

Select Age = Young:

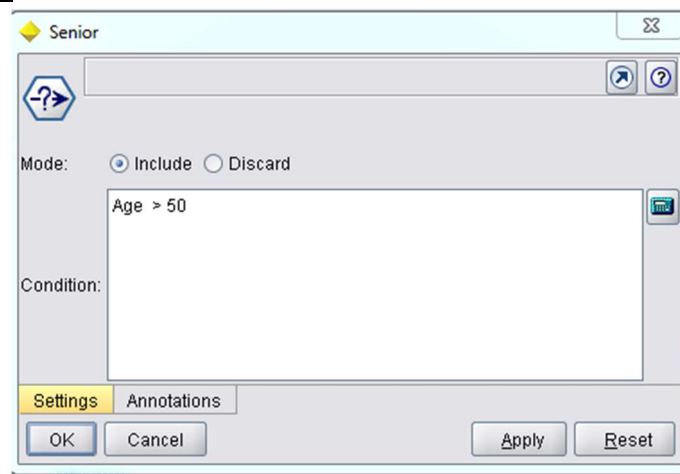


The 'Young' dialog box is shown with the title 'Young'. It contains a section 'Mode:' with radio buttons for 'Include' (selected) and 'Discard'. Below this, there is a text input field containing 'Age <= 30'. A section titled 'Condition:' is empty. At the bottom, there are tabs for 'Settings' (selected) and 'Annotations', and buttons for 'OK', 'Cancel', 'Apply', and 'Reset'.

Select Age = Middle:



Select Age = Senior:



For each Select of age we use Multi Plot as shown in output.

Exercise 3
Using BASKETS1N

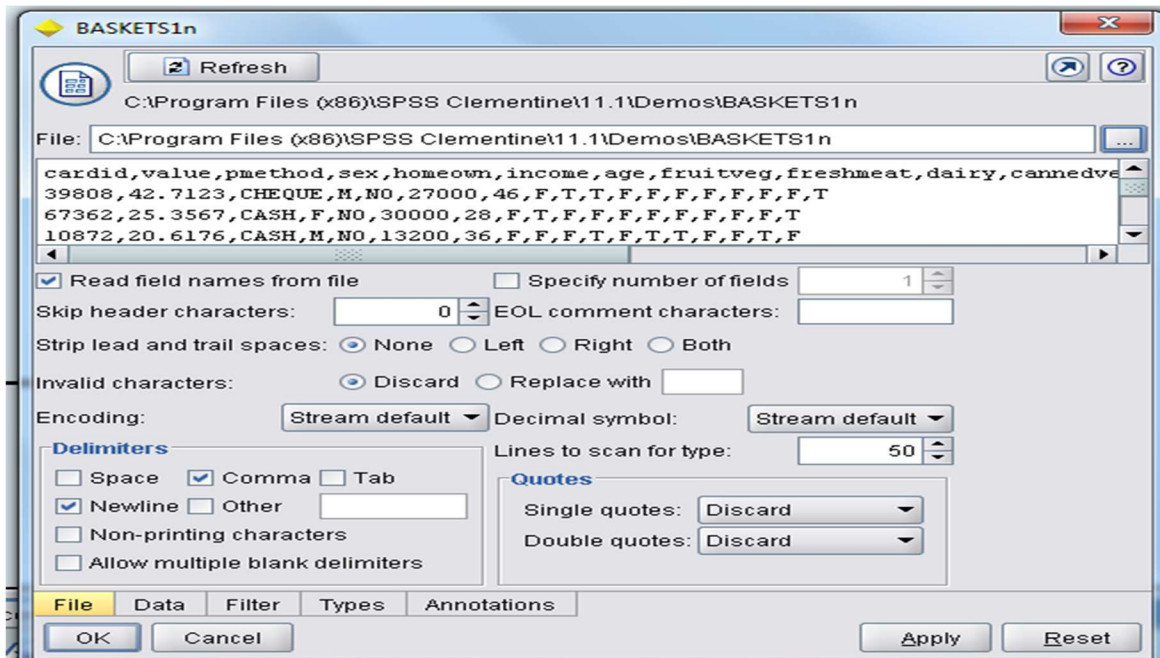
- Find the association rules only for items using Apriori model with minimum support 3% and confidence 90%.
- Compare the GRI and Apriori having support 22% and confidence 90% usage(prepare a sample data set in spreadsheet)
- Determine the Drugs(Drug4n) importance w.r.t Age, Cholesterol and BP and Compare the C5.0 and Neural Net
- Determine the importance of the attributes using K-Means from Drug3n and Drug4n datasets

SOLUTION 3a)

Expected output:



Input data set:



Output dataset:

Consequent	Antecedent	Support %	Confidence %
cannedveg	freshmeat beer frozenmeal	3.0	96.667
frozenmeal	freshmeat beer cannedveg	3.1	93.548
frozenmeal	fish beer cannedveg	4.8	91.667
cannedveg	cannedmeat beer frozenmeal	4.0	90.0

Procedure: The following are the nodes used for this exercise with respective settings.

Type : Using this node we read values and type of each attribute, the non-item attribute are given direction as none and all Item based attributes are given as both input and output to the Apriori Model.

Field	Type	Values	Missing	Check	Direction
cardid	Range	[10150,10...		None	None
value	Range	[10.007,4...		None	None
pmethod	Set	CARD,CA...		None	None
sex	Flag	M/F		None	None
homeown	Flag	YES/NO		None	None
income	Range	[10200,30...		None	None
age	Range	[16,50]		None	None
fruitveg	Flag	T/F		None	Both
freshmeat	Flag	T/F		None	Both

Model : The below diagram represents the settings for the Apriori Model.

8 fields

Model name: ☒ Auto ☐ Custom

☒ Use partitioned data

Minimum antecedent support (%):

Minimum rule confidence (%):

Maximum number of antecedents:

☒ Only true values for flags

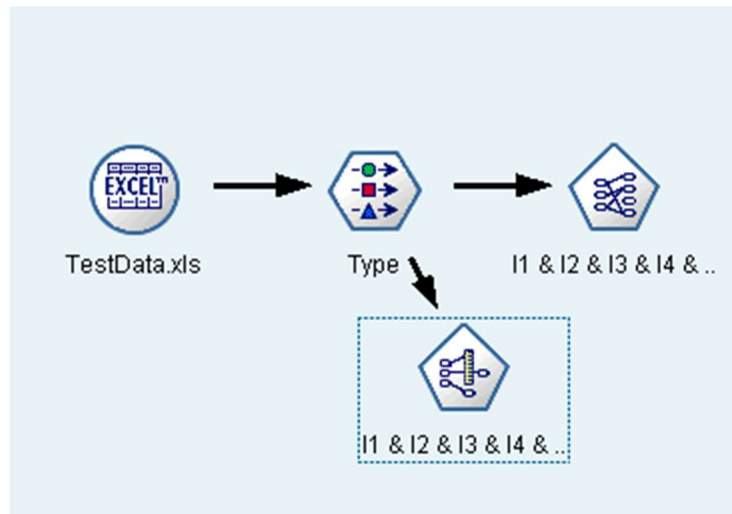
Optimize: ☒ Speed ☐ Memory

Fields Model Expert Annotations

OK Execute Cancel Apply Reset

As we execute this the Apriori Model is build, one we browse the model we can see the resultant rules as output.

SOLUTION 3b)
Expected output:



Input data set: An Excel file is prepared as shown below

TID	I1	I2	I3	I4	I5
T100	T	T	F	F	T
T200	F	T	F	T	F
T300	F	T	T	F	F
T400	T	F	T	T	F
T500	T	F	T	F	F
T600	F	T	T	F	F
T700	T	F	T	F	F
T800	T	T	T	F	T
T900	T	T	T	F	F

TestData.xls

Refresh

C:\Users\TCS\Desktop\TestData.xls

Import file: C:\Users\TCS\Desktop\TestData.xls

☐ Use Named Range

Worksheet: ☒ Index 0 ☐ Name

Data range: ☒ First non-blank row Blank rows: Stop reading ☐ Explicit range

☒ First row contains field names

Data Filter Types Annotations

OK Cancel Apply Reset

Output dataset:

The below outputs shows that there is no difference in rules generated from the Apriori and GRI (Generalized **Rule** Induction), but the order is changed. In Apriori, first low level frequent item sets rules are generated and subsequently the next level frequent itemset rules. Whereas in GRI, the rules are generated on Items i.e. first largest rule then smallest rule size for one item, then the same for subsequent items.

I1 & I2 & I3 & I4 & I5

File Generate

Sort by: Confidence % 4 of 4

Consequent	Antecedent	Support %	Confidence %
I2	I5	22.222	100.0
I1	I5	22.222	100.0
I1	I2	22.222	100.0
I2	I5	22.222	100.0
	I1		

Model Summary Annotations

OK

I1 & I2 & I3 & I4 & I5

File Generate

Sort by: Confidence % 4 of 4

Consequent	Antecedent	Support %	Confidence %
I1	I2	22.22	100.0
I1	I5	22.22	100.0
I2	I1	22.22	100.0
I2	I5	22.22	100.0
	I5		

Model Summary Annotations

OK

Procedure: The following are the nodes used for this exercise with respective settings.

Type

Read Values Clear Values Clear All Values

Field	Type	Values	Missing	Check	Direction
TID	Set	T100,T200...		None	None
I1	Flag	T/F		None	Both
I2	Flag	T/F		None	Both
I3	Flag	T/F		None	Both
I4	Flag	T/F		None	Both
I5	Flag	T/F		None	Both

☒ View current fields ☐ View unused field settings

Types Format Annotations

OK Cancel Apply Reset

Settings for Apriori

I1 & I2 & I3 & I4 & I5

Model name: ☒ Auto ☐ Custom

☒ Use partitioned data

Minimum antecedent support (%): 22.0

Minimum rule confidence (%): 90.0

Maximum number of antecedents: 4

☒ Only true values for flags

Optimize: ☒ Speed ☐ Memory

Fields Model Expert Annotations

OK Execute Cancel Apply Reset

Settings for GRI

I1 & I2 & I3 & I4 & I5

Model name: ☒ Auto ☐ Custom

☒ Use partitioned data

Minimum antecedent support (%):  

Minimum rule confidence (%):  

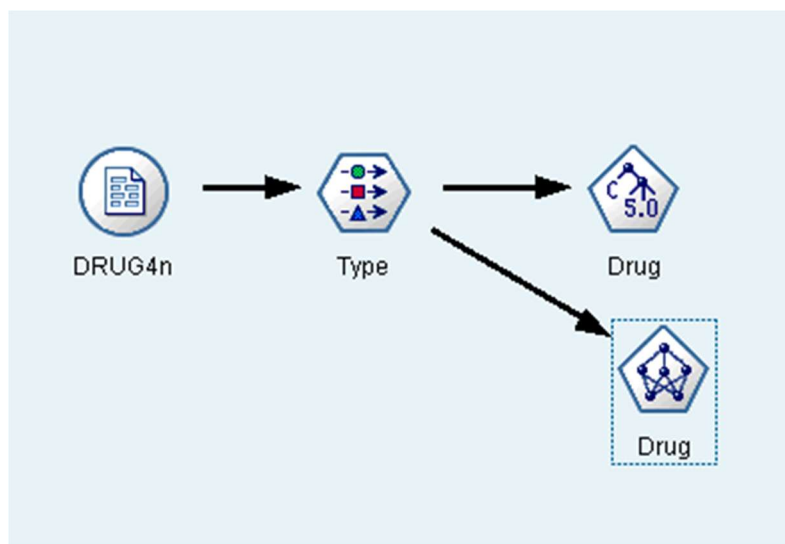
Maximum number of antecedents:  

Maximum number of rules:  

☒ Only true values for flags

Fields **Model** Annotations

SOLUTION 3c)
Expected output:



Input data set:

DRUG4n

Refresh

C:\Program Files (x86)\SPSS Clementine\11.1\Demos\DRUG4n

File: C:\Program Files (x86)\SPSS Clementine\11.1\Demos\DRUG4n

Age, Sex, BP, Cholesterol, Na, K, Drug
 48, F, HIGH, NORMAL, 0.692623, 0.055369, drugA
 15, M, NORMAL, HIGH, 0.678247, 0.040851, drugY
 37, M, HIGH, NORMAL, 0.538192, 0.06978, drugA

☒ Read field names from file ☐ Specify number of fields: 1

Skip header characters: 0 EOL comment characters:

Strip lead and trail spaces: ☒ None ☐ Left ☐ Right ☐ Both

Invalid characters: ☒ Discard ☐ Replace with

Encoding: Stream default Decimal symbol: Stream default

Lines to scan for type: 50

Delimiters

☐ Space ☒ Comma ☐ Tab
☒ Newline ☐ Other
☐ Non-printing characters
☐ Allow multiple blank delimiters

Quotes

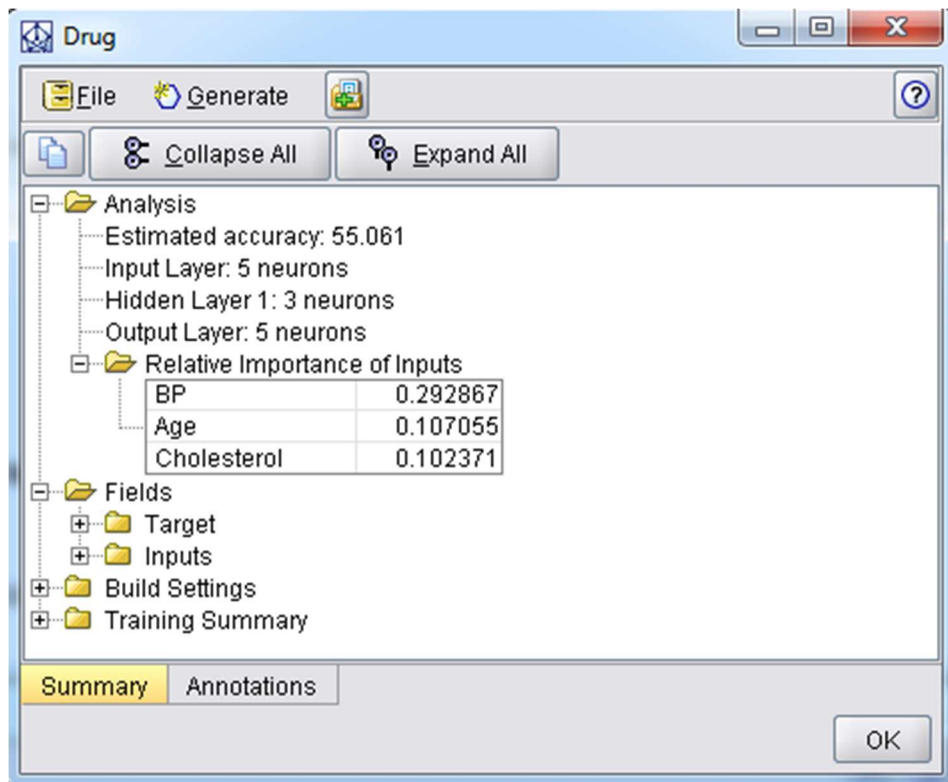
Single quotes: Discard
 Double quotes: Discard

File Data Filter Types Annotations

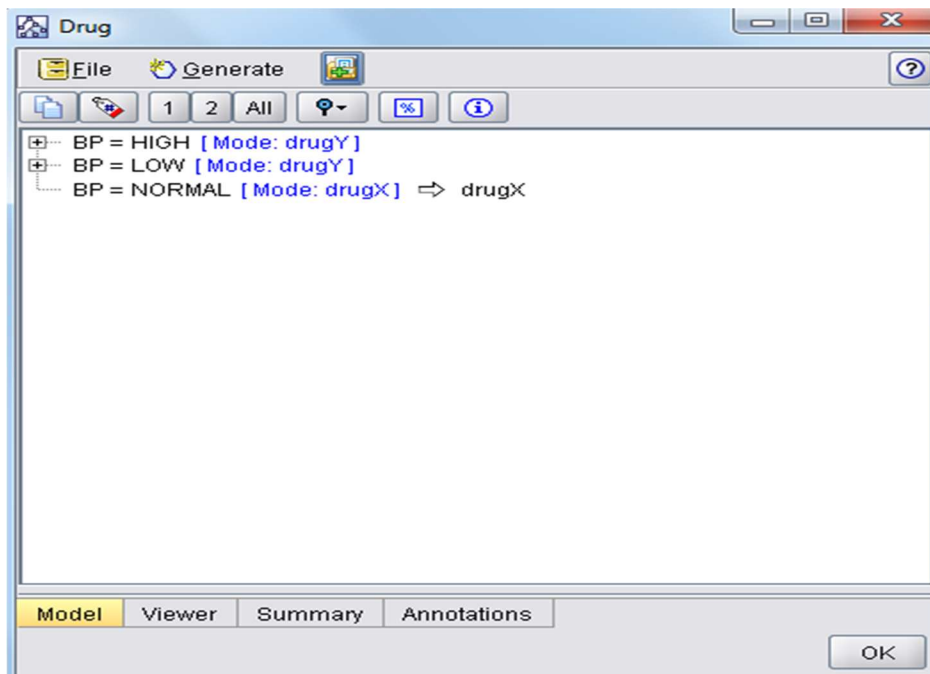
OK Cancel Apply Reset

Output dataset:

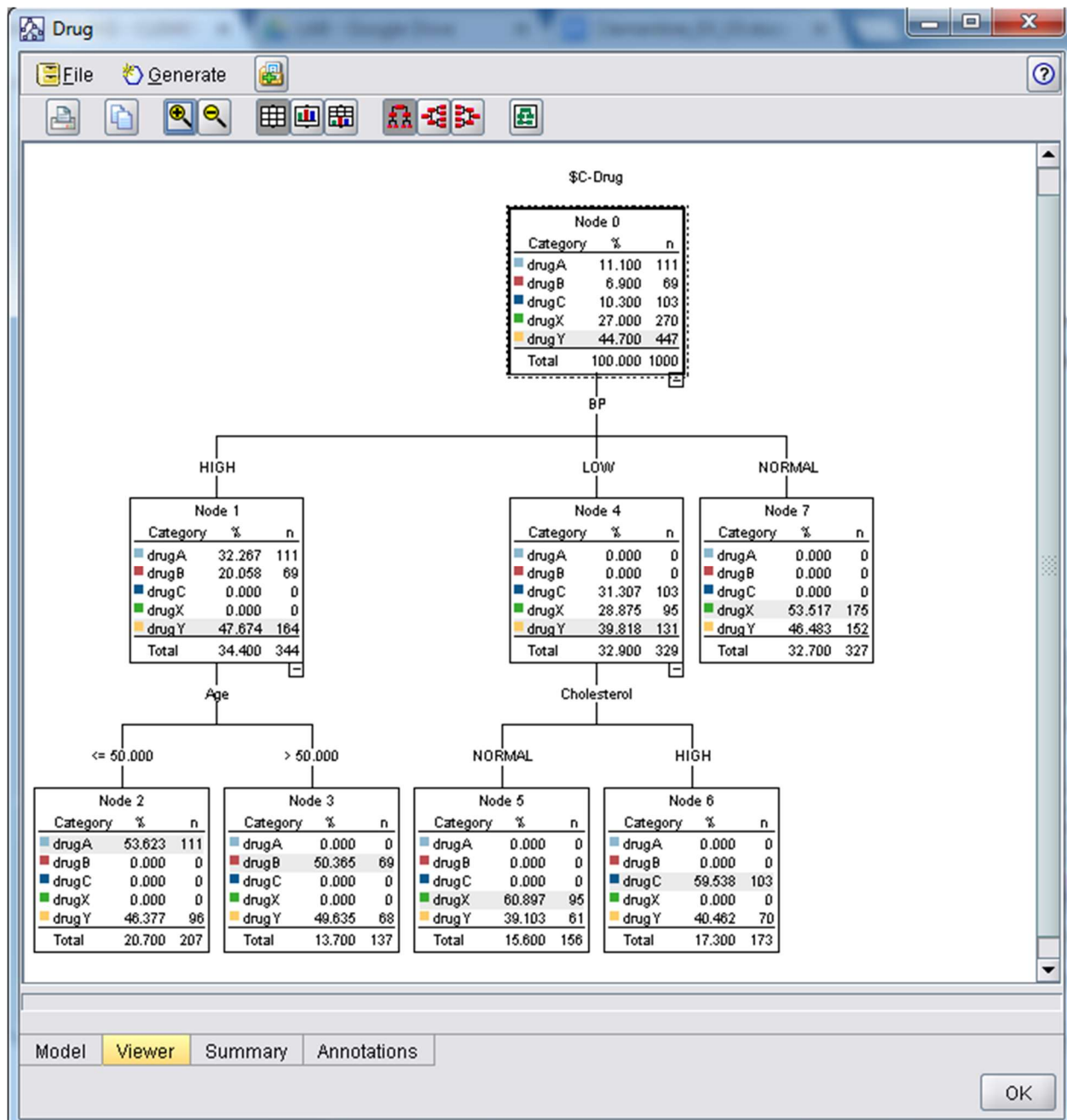
Output of Neural Net Model



Output of C5.0 Model



Output of C5.0 Model viewer which is shown in tree format



Comparison: Attribute Importance is as follows

Neural net Model : BP- 0.2928, Age- 0.1070, Cholestrol- 0.1023

C 5.0 : 2 Level Decision Tree is prepared with BP as root attribute (level 1) with Age and Cholesterol Attributes at level 2

** As a result Neural Net and C 5.0 Models are giving the same information.

Procedure: The following are the nodes used for this exercise with respective settings.

Type

Field	Type	Values	Missing	Check	Direction
Age	Range	[15,74]		None	In
Sex	Flag	M/F		None	In
BP	Set	HIGH,LO...		None	In
Cholesterol	Flag	NORMAL...		None	In
Na	Range	[0.500517,...]		None	In
K	Range	[0.020152,...]		None	In
Drug	Set	drugA,dru...		None	In

☒ View current fields
 ☐ View unused field settings

Types Format Annotations

Settings for C5.0 Model

Drug

Model name: ☒ Auto ☐ Custom

☒ Use partitioned data

Output type: ☒ Decision tree ☐ Rule set

☐ Group symbolics
 ☐ Use boosting Number of trials:

☐ Cross-validate Number of folds:

Mode: ☒ Simple ☐ Expert

Favor: ☒ Accuracy ☐ Generality

Expected noise (%):

Fields **Model** Costs Annotations

Settings for Neural Net Model

Drug

Model name: ☒ Auto ☐ Custom

☒ Use partitioned data

Method:

☒ Prevent overtraining Sample %:

☐ Set random seed Seed:

Stop on: ☒ Default

☐ Accuracy (%)

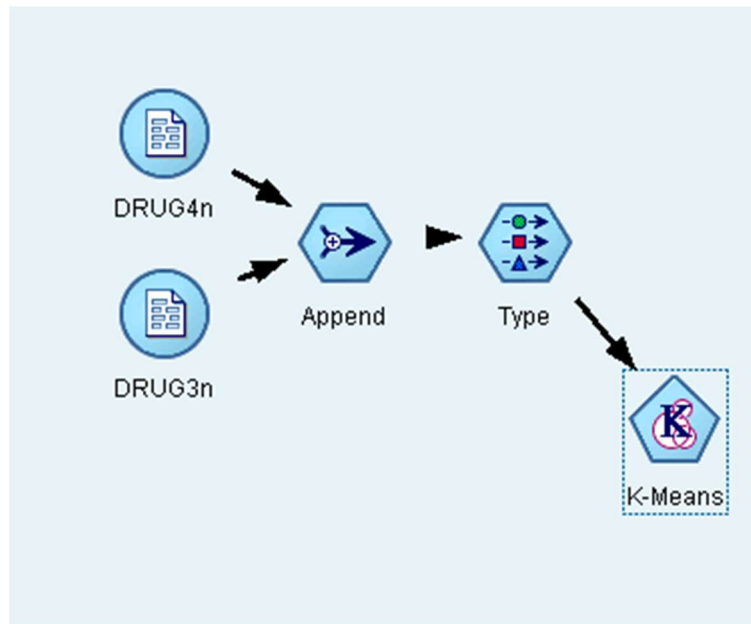
☐ Cycles

☐ Time (mins)

Optimize: ☐ Speed ☒ Memory

Fields **Model** Options Expert Annotations

SOLUTION 3d)
Expected output:



Input data set:

DRUG4n

Refresh

C:\Program Files (x86)\SPSS Clementine\11.1\Demos\DRUG4n

File: C:\Program Files (x86)\SPSS Clementine\11.1\Demos\DRUG4n

Age, Sex, BP, Cholesterol, Na, K, Drug
 48, F, HIGH, NORMAL, 0.692623, 0.055369, drugA
 15, M, NORMAL, HIGH, 0.678247, 0.040851, drugY
 37, M, HIGH, NORMAL, 0.538192, 0.06978, drugA

☒ Read field names from file ☐ Specify number of fields

Skip header characters: EOL comment characters:

Strip lead and trail spaces: ☒ None ☐ Left ☐ Right ☐ Both

Invalid characters: ☒ Discard ☐ Replace with

Encoding: Decimal symbol:

Delimiters

☐ Space ☒ Comma ☐ Tab
☒ Newline ☐ Other
☐ Non-printing characters
☐ Allow multiple blank delimiters

Quotes

Single quotes:
 Double quotes:

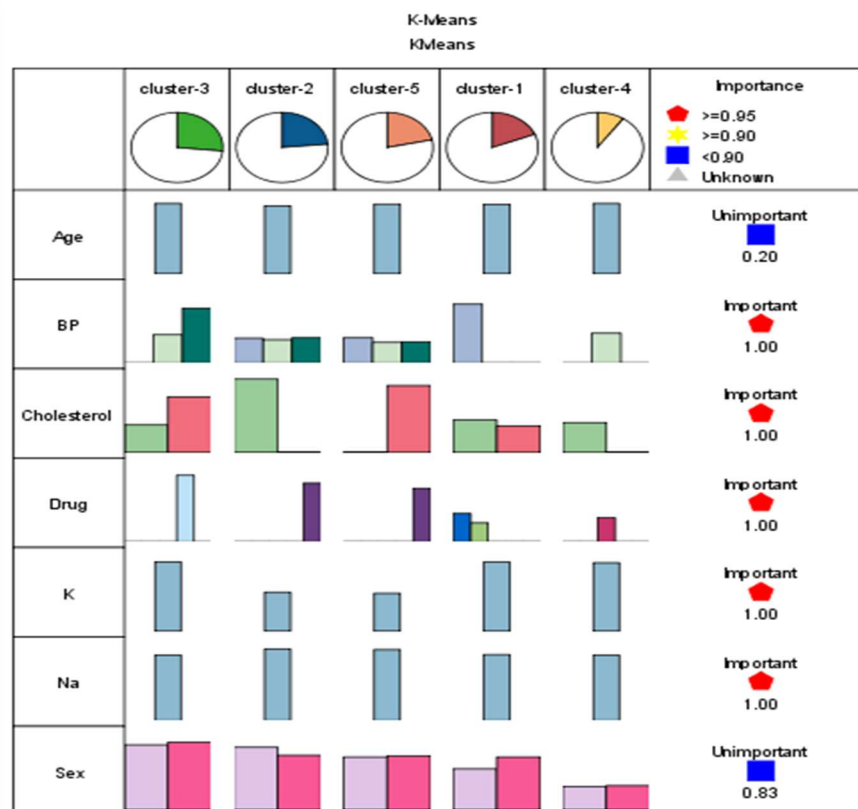
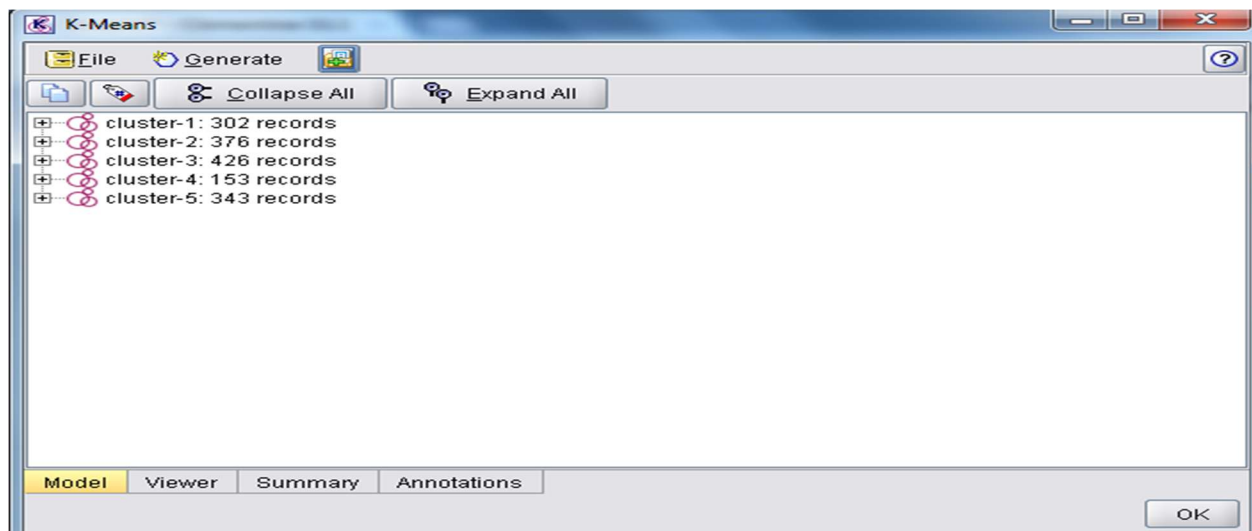
Lines to scan for type:

File Data Filter Types Annotations

OK Cancel Apply Reset

Output dataset:

The following output shows which attributes are important. (The unimportant attributes are Sex and Age)



Procedure: The following are the nodes used for this exercise with respective settings.
Append

Append

Append 2 datasets

Match fields by: ☐ Position ☒ Name ☐ Match case

Preview of field matches and structure

Output Field	1[DRUG4n:DRUG4n]	2[DRUG3n:DRUG3n]
Age	Age	Age
Sex	Sex	Sex
BP	BP	BP
Cholesterol	Cholesterol	Cholesterol
Na	Na	Na
K	K	K
Drug	Drug	Drug

Include fields from: ☒ Main dataset only ☐ All datasets

☐ Tag records by including source dataset in field

Inputs **Append** Annotations

OK Cancel Apply Reset

Setting for K-Means

K-Means

Model name: ☒ Auto ☐ Custom

☒ Use partitioned data

Number of clusters:

☐ Generate distance field

☐ Show cluster proximity

Cluster label: ☒ String ☐ Number

Label prefix:

Optimize: ☐ Speed ☒ Memory

Fields **Model** Expert Annotations

OK **Execute** Cancel Apply Reset